

Initial EDA: Wildfire School Closures, Chronic Absenteeism, and Absenteeism by Reason

Setup

This EDA follows one central question:

Where is wildfire disrupting schooling the most in 2017-18 and 2018-19, and do those same counties show higher chronic absenteeism rates?

The analysis breaks that question into five EDA questions:

1. Are wildfires a major driver of instructional disruption?
2. Which counties, districts, and schools have the most wildfire missed student-days?
3. Which counties have the highest chronic absenteeism rates in 2017-18 and 2018-19?
4. Do counties with higher wildfire missed student-days also have higher chronic absenteeism rates?
5. Among affected school-years, how do absence-day patterns and absence reasons differ by wildfire burden?

The closure dataset uses **year** as the ending year of the academic year. For example, 2019 represents the 2018-19 academic year. The absenteeism analyses therefore use CDE files for 2017-18 and 2018-19, matched to closure **year** == 2018 and **year** == 2019. The 2019-20 absenteeism files are intentionally excluded because CDE marks that year unavailable/not valid due to COVID-era disruption.

This EDA uses two CDE attendance datasets. The **Chronic Absenteeism Data** provide county-level chronic absenteeism counts, eligible enrollment, and rates for the main county comparison. The **Absenteeism by Reason Data** provide school-year absence days and reason composition, which are used as supporting detail rather than as chronic absenteeism rates.

The primary wildfire burden metric is **wildfire missed student-days**, calculated as wildfire closure days multiplied by school-year enrollment. This metric is used because it reflects both the duration of disruption and the number of students exposed. Raw wildfire days and wildfire

share of total closure days remain useful supporting measures, but missed student-days is the main scale for comparing burden across counties, districts, schools, and matched absenteeism records.

days_quarantine and days_safety_other appear to be subcategories of student safety in the codebook, so they are not included in the main top-level closure-cause totals below to avoid double counting.

```
library(tidyverse)
library(janitor)
library(scales)

theme_set(theme_minimal(base_size = 12))

to_number <- function(x) {
  readr::parse_number(as.character(x), na = c("", "NA", "N/A", "NULL", "*"))
}

safe_divide <- function(num, den) {
  if_else(!is.na(den) & den > 0, num / den, NA_real_)
}

clean_cds_code <- function(x) {
  digits <- str_remove_all(coalesce(as.character(x), ""), "\\D")
  if_else(digits == "", NA_character_, str_pad(digits, width = 14, side = "left", pad = "0"))
}

pad_code <- function(x, width) {
  digits <- str_remove_all(coalesce(as.character(x), ""), "\\D")
  if_else(digits == "", NA_character_, str_pad(digits, width = width, side = "left", pad = "0"))
}

year_breaks <- function(x) {
  seq(min(x, na.rm = TRUE), max(x, na.rm = TRUE), by = 2)
}

analysis_years <- c(2018, 2019)
analysis_year_labels <- c("2017-18", "2018-19")

cause_palette <- c(
  "Wildfire" = "#C7432B",
  "Natural disasters/weather" = "#3F7D8C",
  "Infrastructure" = "#D0A03A",
```

```

    "Student safety" = "#5E6E46",
    "Other" = "#7D7A78"
  )

  burden_palette <- c(
    "No wildfire" = "#BFC3C7",
    "Affected: lower 75%" = "#E7A15A",
    "Affected: top 25%" = "#C7432B"
  )

```

Load closure data

```

data_file <- "annual_school_closures_standardized_test_scores_ca_2003_2019.csv"
data_path_candidates <- c(
  file.path("../data/raw/school_fire", data_file),
  file.path("data/raw/school_fire", data_file)
)

data_path <- data_path_candidates[file.exists(data_path_candidates)][1]
if (is.na(data_path)) {
  stop("Could not find the closure CSV. Check the relative path from this Quarto file.")
}

df <- readr::read_csv(
  data_path,
  col_types = readr::cols(
    cds_code = readr::col_character(),
    .default = readr::col_guess()
  ),
  show_col_types = FALSE
) |>
  clean_names()

# The provided column listing uses days_wildfires, while the codebook uses
# Days.Wildfire. Standardize to singular so downstream code is consistent.
if ("days_wildfires" %in% names(df) && !"days_wildfire" %in% names(df)) {
  df <- df |>
    rename(days_wildfire = days_wildfires)
}

```

```

primary_cause_vars <- c(
  "days_wildfire",
  "days_natural_disasters_weather",
  "days_infrastructure",
  "days_student_safety",
  "days_other"
)

closure_numeric_vars <- c(
  primary_cause_vars,
  "days_quarantine",
  "days_safety_other",
  "days_total",
  "total_enrollment"
)

df <- df |>
  mutate(
    year = as.integer(year),
    closure_year = year,
    academic_year = case_when(
      !is.na(year) ~ str_c(year - 1, "-", str_sub(as.character(year), 3, 4)),
      TRUE ~ NA_character_
    ),
    county = str_squish(str_to_title(county)),
    district_name = str_squish(district_name),
    school_site = str_squish(school_site),
    cds_code_raw = cds_code,
    cds_code = clean_cds_code(cds_code),
    across(any_of(closure_numeric_vars), to_number),
    across(any_of(primary_cause_vars), ~ replace_na(.x, 0)),
    days_total = replace_na(days_total, 0),
    total_enrollment = replace_na(total_enrollment, 0),
    wildfire_student_days = days_wildfire * total_enrollment,
    wildfire_share_of_total = safe_divide(days_wildfire, days_total),
    any_wildfire = days_wildfire > 0
  )

cause_lookup <- tibble(
  cause = primary_cause_vars,
  cause_label = factor(
    c(

```

```

      "Wildfire",
      "Natural disasters/weather",
      "Infrastructure",
      "Student safety",
      "Other"
    ),
    levels = c(
      "Wildfire",
      "Natural disasters/weather",
      "Infrastructure",
      "Student safety",
      "Other"
    )
  )
)

glimpse(df)

```

Rows: 203,636

Columns: 116

\$ cds_code	<chr> "01098846987523", "011001~
\$ year	<int> 2015, 2006, 2007, 2008, 2~
\$ county	<chr> "Alameda", "Alameda", "Al~
\$ district_name	<chr> NA, "Alameda County Offic~
\$ school_site	<chr> NA, "FAME Public Charter"~
\$ days_wildfire	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_natural_disasters_weather	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_infrastructure	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_student_safety	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_quarantine	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_safety_other	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_other	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ days_total	<dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
\$ total_enrollment	<dbl> 0, 785, 1143, 1286, 1398,~
\$ percent_female_enrollment	<dbl> NA, 0.5031847, 0.5030621,~
\$ percent_white_enrollment	<dbl> NA, 0.28280255, 0.2852143~
\$ percent_minority_enrollment	<dbl> NA, 0.7171975, 0.7147857,~
\$ percent_asian_enrollment	<dbl> NA, 0.286624204, 0.307961~
\$ percent_hispanic_enrollment	<dbl> NA, 0.08535032, 0.0979877~
\$ percent_black_enrollment	<dbl> NA, 0.14267516, 0.1268591~
\$ percent_esl	<dbl> NA, 0.30828026, 0.4138232~
\$ percent_free_reduced	<dbl> NA, NA, NA, 0.7946578, 0.~

[illegible]

\$ percentage_standard_met_and_above_3_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_3_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_3_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_4_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_4_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_4_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_4_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_4_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_5_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_6_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_7_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_8_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_11_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_exceeded_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_met_and_above_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_nearly_met_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ percentage_standard_not_met_13_english	<dbl> NA, NA, NA, NA, NA, NA, N~
\$ act_avg_score	<dbl> NA, NA, NA, 18.00, NA, NA~
\$ sat_avg_verbal	<dbl> NA, NA, NA, NA, NA, 561.8~
\$ sat_avg_math	<dbl> NA, NA, NA, NA, NA, 576.4~
\$ sat_avg_writing_new	<dbl> NA, NA, NA, NA, NA, 537.2~
\$ sat_avg_total_old	<dbl> NA, NA, NA, NA, NA, 1138.~

```

$ sat_avg_total_new          <dbl> NA, NA, NA, NA, NA, 1675.~
$ percent_ap_345            <dbl> NA, NA, NA, NA, NA, NA, N~
$ closure_year               <int> 2015, 2006, 2007, 2008, 2~
$ academic_year              <chr> "2014-15", "2005-06", "20~
$ cds_code_raw               <chr> "\\01098846987523", "\\01~
$ wildfire_student_days      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ wildfire_share_of_total    <dbl> NA, NA, NA, NA, NA, NA, N~
$ any_wildfire               <lgl> FALSE, FALSE, FALSE, FALS~

```

Closure data quality checks

```

df |>
  mutate(
    primary_cause_sum = rowSums(pick(all_of(primary_cause_vars)), na.rm = TRUE),
    total_minus_primary_causes = days_total - primary_cause_sum
  ) |>
  count(total_minus_primary_causes, sort = TRUE) |>
  slice_head(n = 20)

```

```

# A tibble: 1 x 2
  total_minus_primary_causes      n
      <dbl>   <int>
1             0 203636

```

```

df |>
  mutate(
    student_safety_subtotal = coalesce(days_quarantine, 0) + coalesce(days_safety_other, 0),
    student_safety_difference = days_student_safety - student_safety_subtotal
  ) |>
  count(student_safety_difference, sort = TRUE) |>
  slice_head(n = 20)

```

```

# A tibble: 1 x 2
  student_safety_difference      n
      <dbl>   <int>
1             0 203636

```



```
df |>
  count(cds_code, closure_year, sort = TRUE) |>
  filter(n > 1) |>
  slice_head(n = 20)
```

```
# A tibble: 2 x 3
  cds_code      closure_year      n
  <chr>          <int> <int>
1 31103143130093      2006      2
2 33752423331063      2005      2
```

Step 1: Is wildfire a major driver of instructional disruption?

```
statewide_by_year <- df |>
  group_by(year) |>
  summarise(
    schools = n_distinct(cds_code),
    schools_with_any_closure = n_distinct(cds_code[days_total > 0]),
    total_enrollment = sum(total_enrollment, na.rm = TRUE),
    across(all_of(primary_cause_vars), ~ sum(.x, na.rm = TRUE)),
    days_total = sum(days_total, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(
    primary_cause_sum = rowSums(pick(all_of(primary_cause_vars)), na.rm = TRUE),
    wildfire_share_of_total_days = safe_divide(days_wildfire, days_total),
    wildfire_share_of_primary_causes = safe_divide(days_wildfire, primary_cause_sum),
    focus_year = year %in% analysis_years
  )

statewide_long <- statewide_by_year |>
  select(year, all_of(primary_cause_vars)) |>
  pivot_longer(
    cols = all_of(primary_cause_vars),
    names_to = "cause",
    values_to = "closure_days"
  ) |>
  left_join(cause_lookup, by = "cause")

statewide_by_year |>
```

```

arrange(desc(days_wildfire)) |>
select(
  year,
  schools,
  days_wildfire,
  days_total,
  wildfire_share_of_total_days,
  wildfire_share_of_primary_causes
) |>
slice_head(n = 10)

```

```

# A tibble: 10 x 6
  year schools days_wildfire days_total wildfire_share_of_total_days
  <int>   <int>         <dbl>     <dbl>                <dbl>
1  2018   11677         5860       6415                0.913
2  2008   12249         5354       5974                0.896
3  2004   12072         4448       4648                0.957
4  2019   11581         3893       4898                0.795
5  2016   11572          509       1808                0.282
6  2017   11769          451       2194                0.206
7  2014   11555          297        434                0.684
8  2009   12185          240        970                0.247
9  2010   12109          161       1052                0.153
10 2015   13077           65       1049                0.0620
# i 1 more variable: wildfire_share_of_primary_causes <dbl>

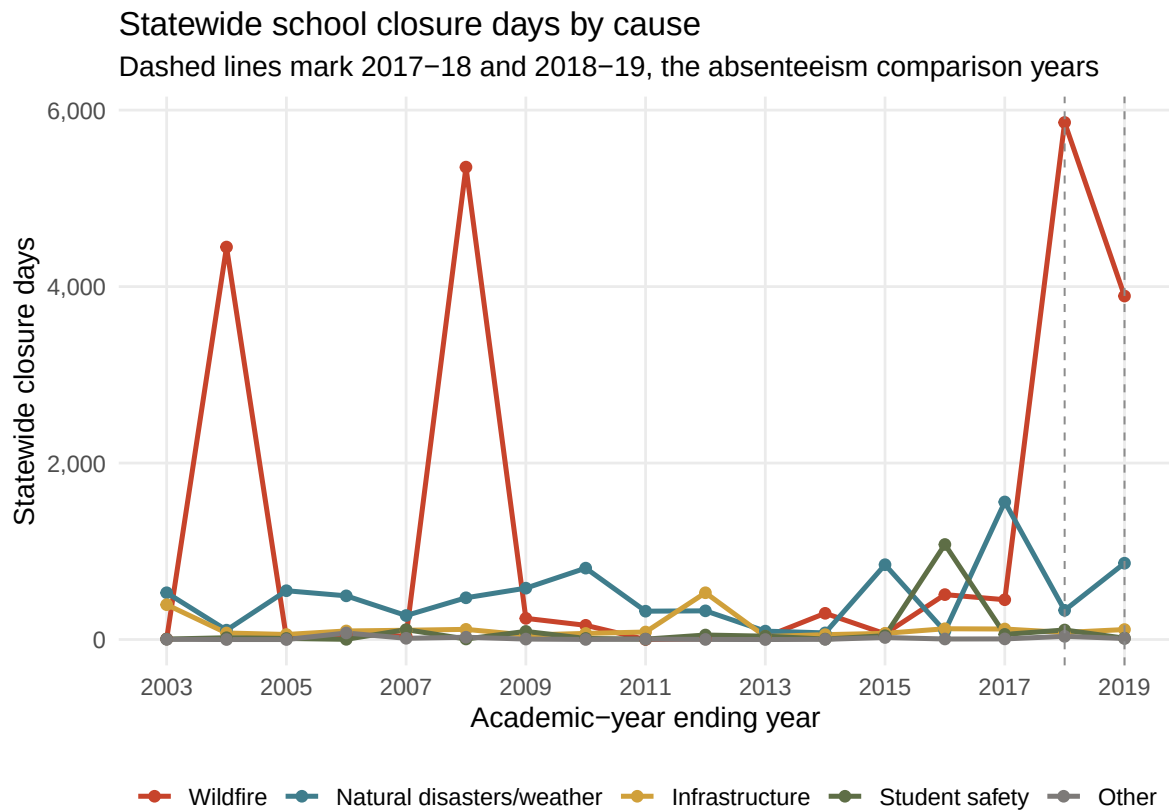
```

```

ggplot(
  statewide_long,
  aes(x = year, y = closure_days, color = cause_label, group = cause_label)
) +
  geom_line(linewidth = 1) +
  geom_point(size = 1.8) +
  geom_vline(xintercept = analysis_years, linetype = "dashed", color = "grey55", linewidth =
  scale_color_manual(values = cause_palette, name = NULL) +
  scale_x_continuous(breaks = year_breaks(statewide_long$year)) +
  scale_y_continuous(labels = label_comma()) +
  labs(
    title = "Statewide school closure days by cause",
    subtitle = "Dashed lines mark 2017-18 and 2018-19, the absenteeism comparison years",
    x = "Academic-year ending year",
    y = "Statewide closure days"
  )

```

```
) +
theme(
  legend.position = "bottom",
  panel.grid.minor = element_blank()
)
```



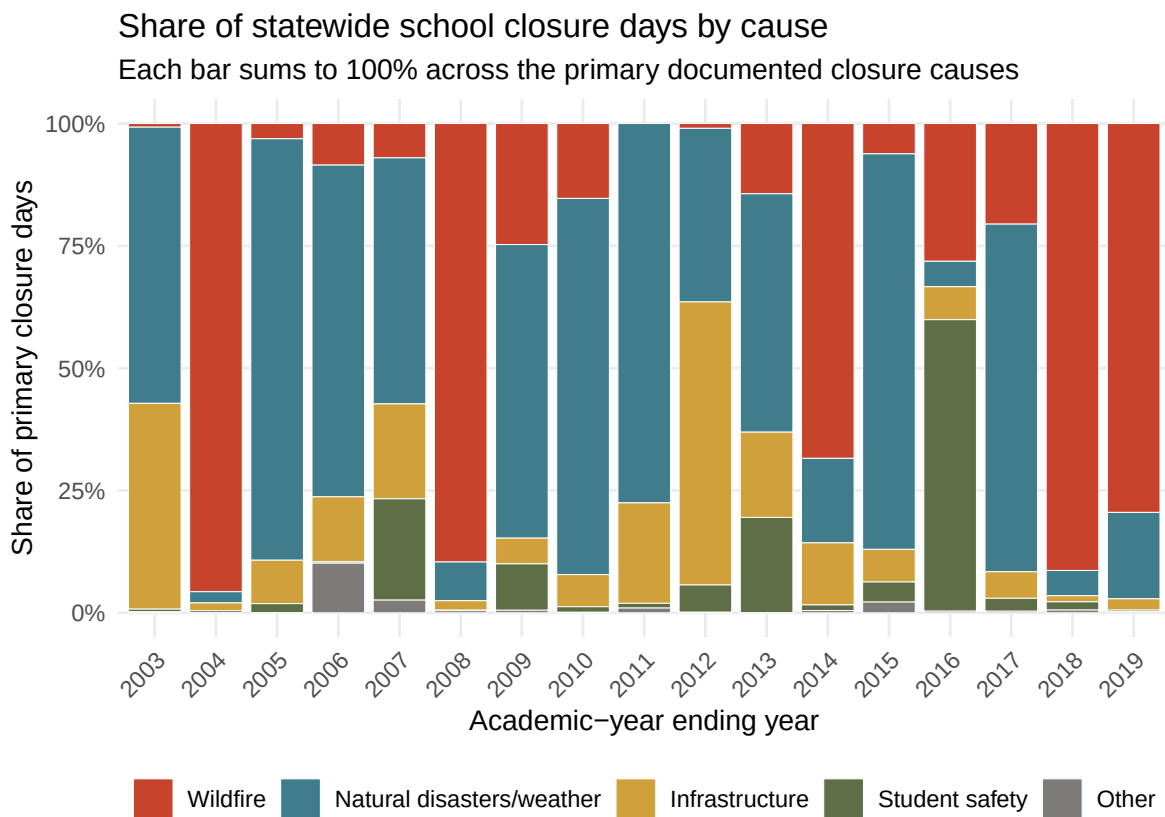
```
statewide_share <- statewide_long |>
  group_by(year) |>
  mutate(
    year_primary_total = sum(closure_days, na.rm = TRUE),
    share = safe_divide(closure_days, year_primary_total)
  ) |>
  ungroup() |>
  filter(!is.na(share))

ggplot(
  statewide_share,
```

```

aes(x = factor(year), y = share, fill = cause_label)
) +
geom_col(width = 0.86, color = "white", linewidth = 0.15) +
scale_fill_manual(values = cause_palette, name = NULL) +
scale_y_continuous(labels = label_percent()) +
labs(
  title = "Share of statewide school closure days by cause",
  subtitle = "Each bar sums to 100% across the primary documented closure causes",
  x = "Academic-year ending year",
  y = "Share of primary closure days"
) +
theme(
  legend.position = "bottom",
  panel.grid.minor = element_blank(),
  axis.text.x = element_text(angle = 45, hjust = 1)
)

```

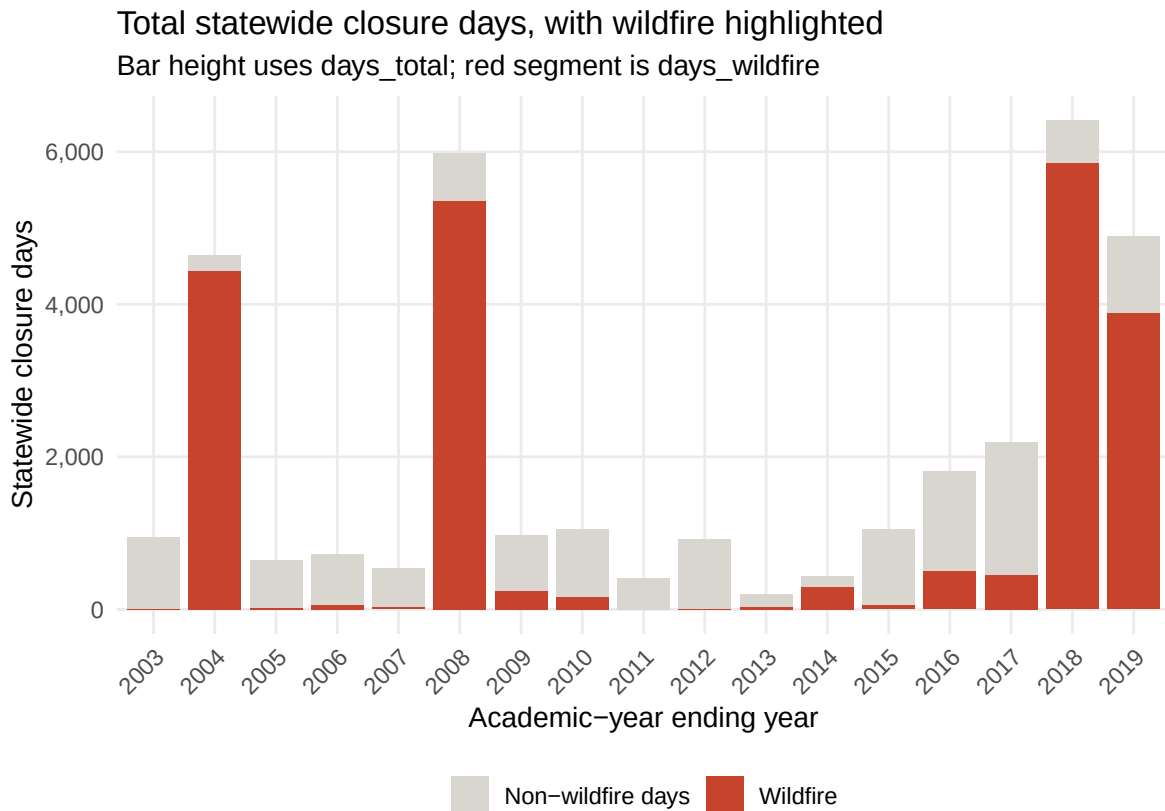


```

total_with_wildfire_component <- statewide_by_year |>
  transmute(
    year,
    Wildfire = days_wildfire,
    `Non-wildfire days` = pmax(days_total - days_wildfire, 0)
  ) |>
  pivot_longer(
    cols = -year,
    names_to = "component",
    values_to = "closure_days"
  ) |>
  mutate(
    component = factor(component, levels = c("Non-wildfire days", "Wildfire"))
  )

ggplot(
  total_with_wildfire_component,
  aes(x = factor(year), y = closure_days, fill = component)
) +
  geom_col(width = 0.86) +
  scale_fill_manual(
    values = c(
      "Non-wildfire days" = "#D9D6CF",
      "Wildfire" = "#C7432B"
    ),
    name = NULL
  ) +
  scale_y_continuous(labels = label_comma()) +
  labs(
    title = "Total statewide closure days, with wildfire highlighted",
    subtitle = "Bar height uses days_total; red segment is days_wildfire",
    x = "Academic-year ending year",
    y = "Statewide closure days"
  ) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 45, hjust = 1)
  )

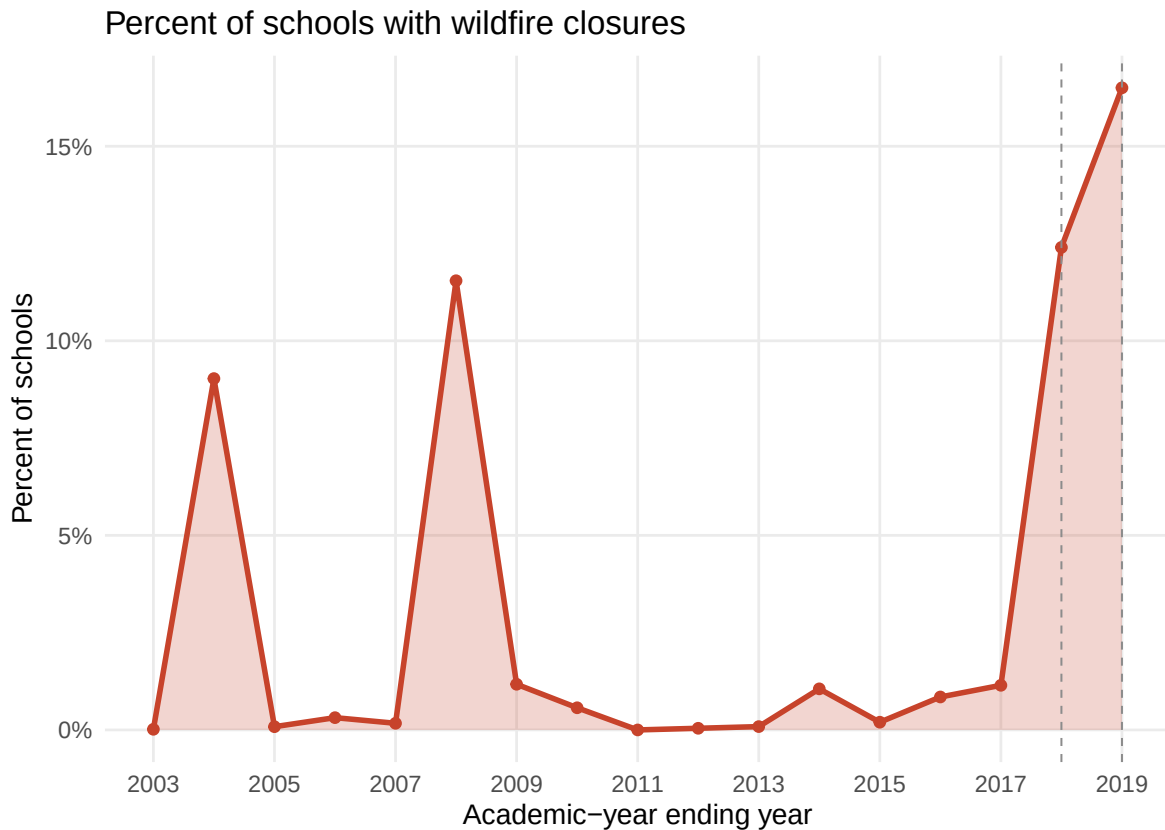
```



```
wildfire_reach_by_year <- df |>
  group_by(year) |>
  summarise(
    schools = n_distinct(cds_code),
    schools_with_wildfire_closure = n_distinct(cds_code[days_wildfire > 0]),
    pct_schools_with_wildfire_closure = safe_divide(schools_with_wildfire_closure, schools),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    .groups = "drop"
  )

ggplot(
  wildfire_reach_by_year,
  aes(x = year, y = pct_schools_with_wildfire_closure)
) +
  geom_area(fill = "#C7432B", alpha = 0.22) +
  geom_line(color = "#C7432B", linewidth = 1.1) +
  geom_point(color = "#C7432B", size = 1.8) +
  geom_vline(xintercept = analysis_years, linetype = "dashed", color = "grey55", linewidth =
```

```
scale_x_continuous(breaks = year_breaks(wildfire_reach_by_year$year)) +
scale_y_continuous(labels = label_percent(), limits = c(0, NA)) +
labs(
  title = "Percent of schools with wildfire closures",
  x = "Academic-year ending year",
  y = "Percent of schools"
) +
theme(panel.grid.minor = element_blank())
```



Step 1 takeaway

Wildfire is a major driver of statewide instructional disruption, especially in peak closure years. The statewide rankings and cause-share plots show that 2018 and 2019 are central to the recent pre-pandemic wildfire story: wildfire closure days account for most statewide closure days in those years. This supports focusing the absenteeism comparison on the matched 2017-18 and 2018-19 school years.

Step 2: Is the burden uneven across counties, districts, and schools?

This section focuses on 2017-18 and 2018-19 to match the absenteeism files.

```
closure_window <- df |>
  filter(year %in% analysis_years)

county_window_burden <- closure_window |>
  group_by(county) |>
  summarise(
    school_year_rows = n(),
    schools = n_distinct(cds_code),
    districts = n_distinct(district_name),
    total_enrollment_school_years = sum(total_enrollment, na.rm = TRUE),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    total_closure_days = sum(days_total, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    avg_wildfire_days_per_school_year = safe_divide(wildfire_days, school_year_rows),
    wildfire_share_of_total = safe_divide(wildfire_days, total_closure_days),
    .groups = "drop"
  ) |>
  filter(!is.na(county), county != "")

district_window_burden <- closure_window |>
  group_by(county, district_name) |>
  summarise(
    school_year_rows = n(),
    schools = n_distinct(cds_code),
    total_enrollment_school_years = sum(total_enrollment, na.rm = TRUE),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    total_closure_days = sum(days_total, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    avg_wildfire_days_per_school_year = safe_divide(wildfire_days, school_year_rows),
    wildfire_share_of_total = safe_divide(wildfire_days, total_closure_days),
    .groups = "drop"
  ) |>
  filter(!is.na(district_name), district_name != "")

school_window_burden <- closure_window |>
  group_by(cds_code, county, district_name, school_site) |>
  summarise(
    school_year_rows = n(),
    years_observed = n_distinct(year),
```



```

total_enrollment_school_years = sum(total_enrollment, na.rm = TRUE),
wildfire_days = sum(days_wildfire, na.rm = TRUE),
total_closure_days = sum(days_total, na.rm = TRUE),
wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
avg_wildfire_days_per_school_year = safe_divide(wildfire_days, school_year_rows),
wildfire_share_of_total = safe_divide(wildfire_days, total_closure_days),
  .groups = "drop"
) |>
filter(!is.na(cds_code))

list(
  counties_with_wildfire = county_window_burden |> filter(wildfire_days > 0) |> nrow(),
  districts_with_wildfire = district_window_burden |> filter(wildfire_days > 0) |> nrow(),
  schools_with_wildfire = school_window_burden |> filter(wildfire_days > 0) |> nrow()
)

```

```

$counties_with_wildfire
[1] 36

```

```

$districts_with_wildfire
[1] 326

```

```

$schools_with_wildfire
[1] 2721

```

County burden rankings

```

top_n_counties <- 20

county_window_burden |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    schools,
    districts,
    wildfire_days,
    wildfire_student_days,
    avg_wildfire_days_per_school_year,
    wildfire_share_of_total
  )

```

```

) |>
slice_head(n = top_n_counties)

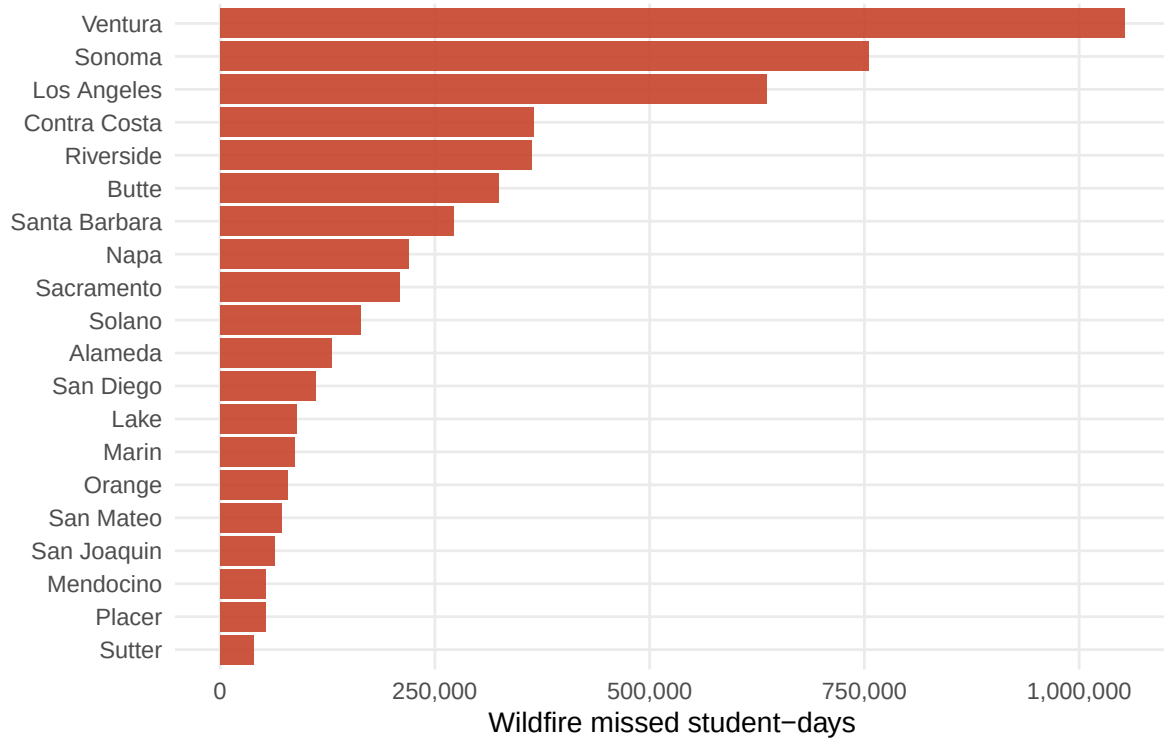
# A tibble: 20 x 7
  county      schools districts wildfire_days wildfire_student_days
  <chr>      <int>    <int>      <dbl>          <dbl>
1 Ventura      256      22      1540      1052437
2 Sonoma       244      43      1805      755475
3 Los Angeles  2508     100       879      635624
4 Contra Costa  322      22       596      364434
5 Riverside    576      27       371      362921
6 Butte        114      16       928      324612
7 Santa Barbara 142      27       570      271895
8 Napa         52       8       447      219721
9 Sacramento   425     17       305      209115
10 Solano      119      8       253      163051
11 Alameda     458     26       178      129436
12 San Diego   878     53       168      111207
13 Lake        46       8       344      88748
14 Marin       108     22       222      86526
15 Orange      693     34       107      78093
16 San Mateo   233     26       125      71285
17 San Joaquin 272     17       100      62862
18 Mendocino   77      15       181      53199
19 Placer      160     19        71      52751
20 Sutter       53     15        83      39253

# i 2 more variables: avg_wildfire_days_per_school_year <dbl>,
#   wildfire_share_of_total <dbl>

county_window_burden |>
  slice_max(wildfire_student_days, n = top_n_counties, with_ties = FALSE) |>
  ggplot(aes(x = wildfire_student_days, y = fct_reorder(county, wildfire_student_days))) +
  geom_col(fill = "#C7432B", alpha = 0.9) +
  scale_x_continuous(labels = label_comma()) +
  labs(
    title = "Counties with the largest wildfire missed student-days",
    subtitle = "2017-18 and 2018-19 only",
    x = "Wildfire missed student-days",
    y = NULL
  ) +
  theme(panel.grid.minor = element_blank())

```

Counties with the largest wildfire missed student-days 2017-18 and 2018-19 only



District burden rankings

```
top_n_districts <- 25

district_window_burden |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    district_name,
    schools,
    wildfire_days,
    wildfire_student_days,
    avg_wildfire_days_per_school_year,
    wildfire_share_of_total
  ) |>
  slice_head(n = top_n_districts)
```

```
# A tibble: 25 x 7
```

	county	district_name	schools	wildfire_days	wildfire_student_days
	<chr>	<chr>	<int>	<dbl>	<dbl>
1	Los Angeles	Los Angeles Unified	1017	789	572155
2	Ventura	Ventura Unified	28	390	251415
3	Napa	Napa Valley Unified	31	319	193670
4	Sonoma	Santa Rosa High	12	203	187754
5	Ventura	Oxnard Union High	13	114	175339
6	Ventura	Conejo Valley Unified	28	207	146813
7	Butte	Chico Unified	30	283	140817
8	Riverside	Corona-Norco Unified	54	139	139754
9	Ventura	Oxnard	21	160	132273
10	Riverside	Lake Elsinore Unified	26	115	106230

```
# i 15 more rows
```

```
# i 2 more variables: avg_wildfire_days_per_school_year <dbl>,
```

```
# wildfire_share_of_total <dbl>
```

```
district_window_burden |>
```

```
  mutate(district_label = str_c(str_trunc(district_name, 42), " (", county, ")")) |>
```

```
  slice_max(wildfire_student_days, n = 18, with_ties = FALSE) |>
```

```
  ggplot(aes(x = wildfire_student_days, y = fct_reorder(district_label, wildfire_student_days,
```

```
    geom_col(fill = "#B95A3C", alpha = 0.9) +
```

```
    scale_x_continuous(labels = label_comma()) +
```

```
    labs(
```

```
      title = "Districts with the largest wildfire missed student-days",
```

```
      subtitle = "2017-18 and 2018-19 only",
```

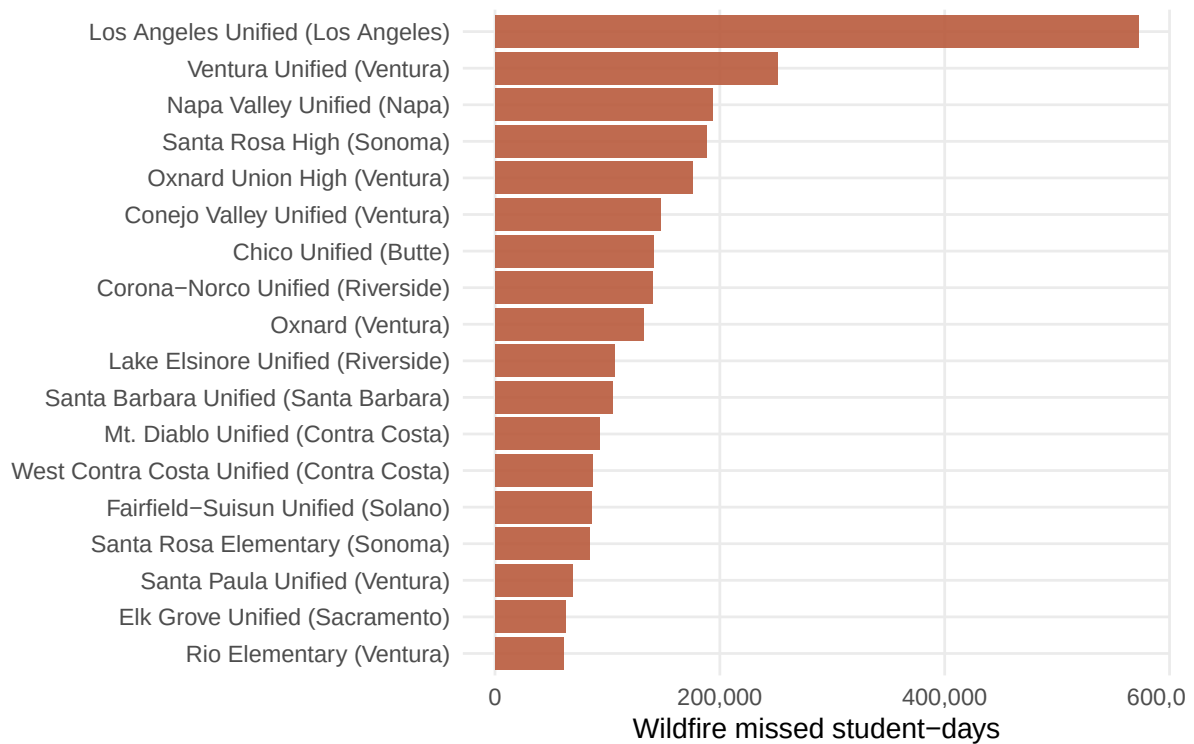
```
      x = "Wildfire missed student-days",
```

```
      y = NULL
```

```
    ) +
```

```
    theme(panel.grid.minor = element_blank())
```

Districts with the largest wildfire missed student-c 2017-18 and 2018-19 only



School burden rankings

```
top_n_schools <- 25

top_wildfire_schools <- school_window_burden |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    district_name,
    school_site,
    years_observed,
    total_enrollment_school_years,
    wildfire_days,
    wildfire_student_days,
    avg_wildfire_days_per_school_year,
    wildfire_share_of_total
  )
```

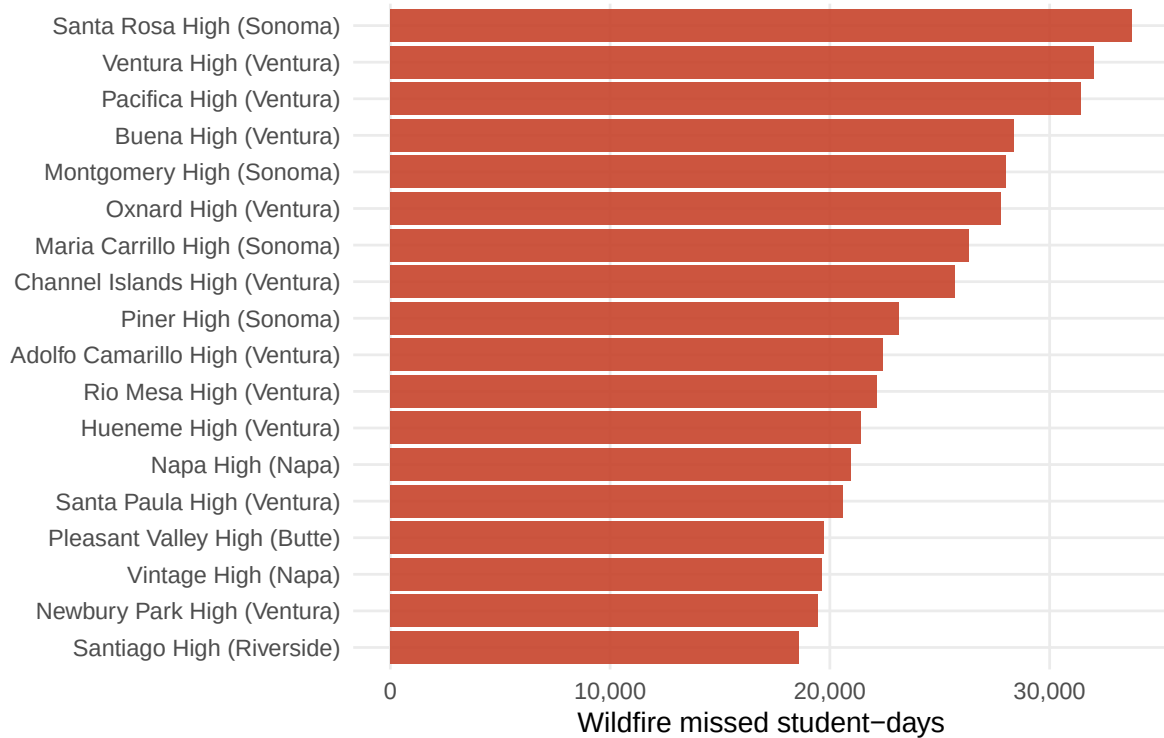
```
) |>
  slice_head(n = top_n_schools)

top_wildfire_schools
```

```
# A tibble: 25 x 9
  county district_name school_site years_observed total_enrollment_sch~1
  <chr>   <chr>         <chr>         <int>         <dbl>
1 Sonoma Santa Rosa High Santa Rosa H~      2      3939
2 Ventura Ventura Unified Ventura High      2      4207
3 Ventura Oxnard Union High Pacifica High      2      6252
4 Ventura Ventura Unified Buena High      2      3739
5 Sonoma Santa Rosa High Montgomery H~      2      3289
6 Ventura Oxnard Union High Oxnard High      2      5515
7 Sonoma Santa Rosa High Maria Carril~      2      3022
8 Ventura Oxnard Union High Channel Isla~      2      5155
9 Sonoma Santa Rosa High Piner High      2      2745
10 Ventura Oxnard Union High Adolfo Camar~      2      4470
# i 15 more rows
# i abbreviated name: 1: total_enrollment_school_years
# i 4 more variables: wildfire_days <dbl>, wildfire_student_days <dbl>,
#   avg_wildfire_days_per_school_year <dbl>, wildfire_share_of_total <dbl>
```

```
school_window_burden |>
  mutate(school_label = str_c(str_trunc(school_site, 42), " (", county, ")")) |>
  slice_max(wildfire_student_days, n = 18, with_ties = FALSE) |>
  ggplot(aes(x = wildfire_student_days, y = fct_reorder(school_label, wildfire_student_days)) +
    geom_col(fill = "#C7432B", alpha = 0.9) +
    scale_x_continuous(labels = label_comma()) +
    labs(
      title = "Schools with the largest wildfire missed student-days",
      subtitle = "2017-18 and 2018-19 only",
      x = "Wildfire missed student-days",
      y = NULL
    ) +
    theme(panel.grid.minor = element_blank()))
```

Schools with the largest wildfire missed student-days 2017-18 and 2018-19 only



Concentration of wildfire burden

```
make_concentration <- function(data, unit_label) {
  data |>
    filter(wildfire_student_days > 0) |>
    arrange(desc(wildfire_student_days)) |>
    mutate(
      rank = row_number(),
      rank_share = rank / n(),
      cumulative_share = cumsum(wildfire_student_days) / sum(wildfire_student_days, na.rm = TRUE),
      unit_type = unit_label
    )
}

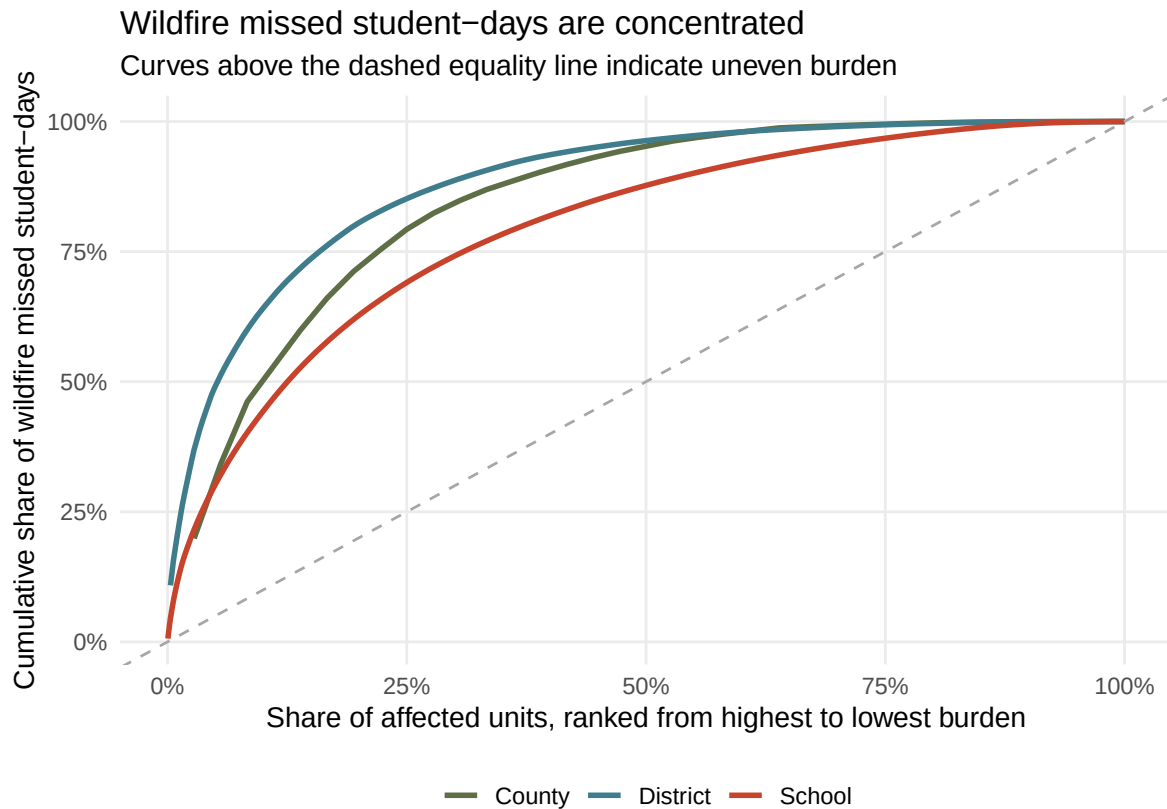
concentration_data <- bind_rows(
  make_concentration(school_window_burden, "School"),
```

```

make_concentration(district_window_burden, "District"),
make_concentration(county_window_burden, "County")
)

ggplot(
  concentration_data,
  aes(x = rank_share, y = cumulative_share, color = unit_type)
) +
  geom_line(linewidth = 1.1) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "grey65") +
  scale_x_continuous(labels = label_percent()) +
  scale_y_continuous(labels = label_percent()) +
  scale_color_manual(values = c("School" = "#C7432B", "District" = "#3F7D8C", "County" = "#5
  labs(
    title = "Wildfire missed student-days are concentrated",
    subtitle = "Curves above the dashed equality line indicate uneven burden",
    x = "Share of affected units, ranked from highest to lowest burden",
    y = "Cumulative share of wildfire missed student-days"
  ) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )
)

```

County maps

```
load_ca_counties <- function() {
  if (!requireNamespace("sf", quietly = TRUE)) {
    message("Package 'sf' is not installed; skipping map chunks.")
    return(NULL)
  }

  county_zip_url <- "https://www2.census.gov/geo/tiger/GENZ2018/shp/cb_2018_us_county_500k.zip"
  zip_path <- tempfile(fileext = ".zip")
  unzip_dir <- tempfile()
  dir.create(unzip_dir)

  tryCatch(
    {
      download.file(county_zip_url, zip_path, mode = "wb", quiet = TRUE)
    }
  )
}
```

```

unzip(zip_path, exdir = unzip_dir)
shp_path <- list.files(unzip_dir, pattern = "\\shp$", full.names = TRUE)[1]

sf::st_read(shp_path, quiet = TRUE) |>
  filter(STATEFP == "06") |>
  sf::st_transform(3310)
},
error = function(e) {
  message("County boundary download/read failed; skipping maps. Error: ", conditionMessage(e))
  NULL
}
)
}

ca_counties <- load_ca_counties()

county_map_data <- if (!is.null(ca_counties)) {
  ca_counties |>
  left_join(county_window_burden, by = c("NAME" = "county"))
} else {
  NULL
}

if (is.null(county_map_data)) {
  tibble(map_status = "County map data unavailable; non-map EDA sections still render.")
} else {
  county_map_data |>
  sf::st_drop_geometry() |>
  summarise(
    counties = n(),
    counties_matched_to_closure_data = sum(!is.na(wildfire_student_days)),
    counties_with_wildfire = sum(coalesce(wildfire_student_days, 0) > 0)
  )
}

```

	counties	counties_matched_to_closure_data	counties_with_wildfire
1	58	58	36

```

if (!is.null(county_map_data)) {
  p1 <- ggplot(county_map_data) +
    geom_sf(aes(fill = wildfire_student_days), color = "white", linewidth = 0.15) +

```

```

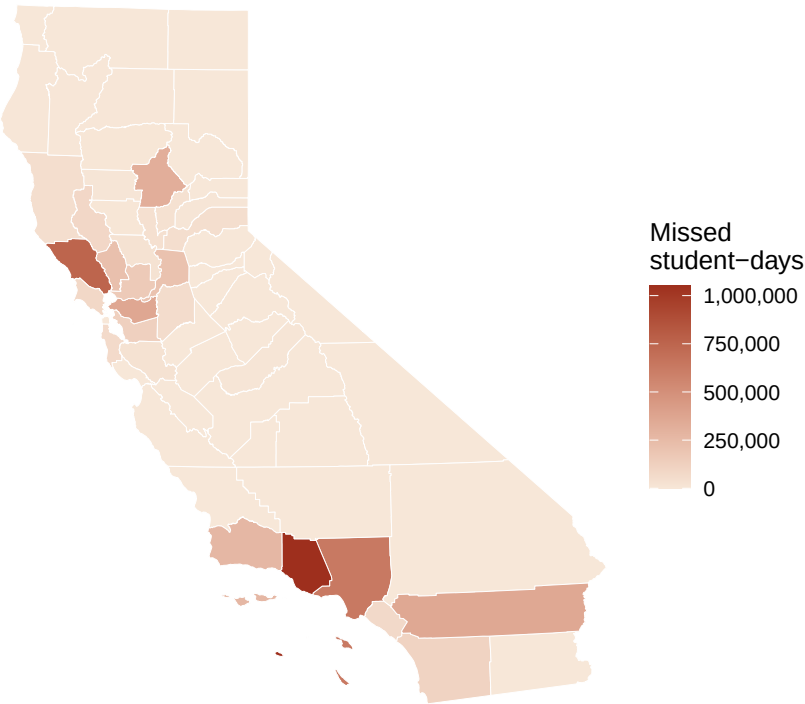
scale_fill_gradient(
  low = "#F7E7D8",
  high = "#9E2F1D",
  labels = label_comma(),
  na.value = "#F1F1F1",
  name = "Missed\nstudent-days"
) +
labs(
  title = "Wildfire missed student-days by county",
  subtitle = "2017-18 and 2018-19"
) +
theme_void() +
theme(legend.position = "right")

p2 <- ggplot(county_map_data) +
  geom_sf(aes(fill = avg_wildfire_days_per_school_year), color = "white", linewidth = 0.15)
scale_fill_gradient(
  low = "#F7E7D8",
  high = "#9E2F1D",
  labels = label_number(accuracy = 0.01),
  na.value = "#F1F1F1",
  name = "Avg days\nper school-year"
) +
labs(
  title = "Average wildfire closure days per school-year by county",
  subtitle = "2017-18 and 2018-19"
) +
theme_void() +
theme(legend.position = "right")

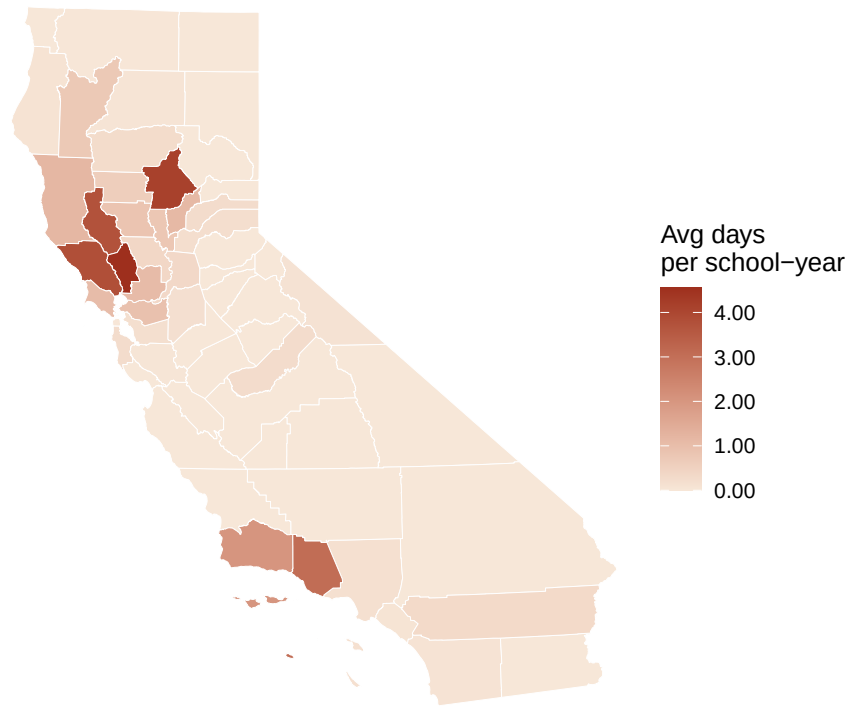
print(p1)
print(p2)
}

```

Wildfire missed student-days by county
2017-18 and 2018-19



Average wildfire closure days per school-year by county 2017-18 and 2018-19



Step 2 takeaway

Wildfire burden is widespread but uneven. In the 2017-18 and 2018-19 window, many counties, districts, and schools have some wildfire closure burden, but missed student-days are concentrated in a much smaller set of places. District counts can vary depending on whether districts are counted as county-district pairs or by district name alone; this notebook uses county-district pairs for the district burden tables because district names are not guaranteed to be unique statewide.

Step 3: Which counties have the highest chronic absenteeism rates?

Load CDE Chronic Absenteeism data

CDE file source: <https://www.cde.ca.gov/ds/ad/fsabd.asp>

The chronic absenteeism files provide the main county-level outcome for this EDA. The analysis uses county total records (`aggregate_level == "C"` and `reporting_category == "TA"`) so

rates are not duplicated across student-group rows. Older chronic absenteeism files do not include a `dass` field, so the reader below harmonizes that field to "All" when absent.

```
chronic_absenteeism_numeric_vars <- c(
  "chronic_absenteeism_eligible_cumulative_enrollment",
  "chronic_absenteeism_count",
  "chronic_absenteeism_rate_reported"
)

resolve_attendance_source <- function(file_name, url) {
  source_candidates <- c(
    file.path("../", "data", "raw", "cde_attendance", file_name),
    file.path("data", "raw", "cde_attendance", file_name)
  )

  local_source <- source_candidates[file.exists(source_candidates)][1]
  if (!is.na(local_source)) {
    return(local_source)
  }

  url
}

empty_chronic_absenteeism <- function(closure_year, source, status) {
  tibble(
    academic_year = character(),
    aggregate_level = character(),
    county_code = character(),
    district_code = character(),
    school_code = character(),
    county_name = character(),
    district_name = character(),
    school_name = character(),
    charter = character(),
    dass = character(),
    reporting_category = character(),
    chronic_absenteeism_eligible_cumulative_enrollment = character(),
    chronic_absenteeism_count = character(),
    chronic_absenteeism_rate_reported = character(),
    closure_year = integer(),
    source_url = character(),
    chronic_absenteeism_read_status = character()
  ) |>
```

```

    add_row(
      closure_year = closure_year,
      source_url = source,
      chronic_absenteeism_read_status = status
    )
  }

standardize_chronic_absenteeism_names <- function(data) {
  if ("charter_yn" %in% names(data) && !"charter" %in% names(data)) {
    data <- data |> rename(charter = charter_yn)
  }

  if (!"dass" %in% names(data)) {
    data <- data |> mutate(dass = "All")
  }

  eligible_col <- names(data)[str_detect(names(data), "^chronic_absenteeism_eligible")]
  if (
    !"chronic_absenteeism_eligible_cumulative_enrollment" %in% names(data) &&
    length(eligible_col) > 0
  ) {
    data <- data |>
      rename(chronic_absenteeism_eligible_cumulative_enrollment = all_of(eligible_col[1]))
  }

  if ("chronic_absenteeism_rate" %in% names(data)) {
    data <- data |> rename(chronic_absenteeism_rate_reported = chronic_absenteeism_rate)
  }

  data
}

read_tsv_source <- function(source) {
  if (file.exists(source)) {
    return(
      readr::read_tsv(
        source,
        col_types = readr::cols(.default = readr::col_character()),
        na = c("", "NA"),
        show_col_types = FALSE
      )
    )
  }
}

```

```

}

lines <- readLines(source, warn = FALSE)
lines <- iconv(lines, from = "", to = "UTF-8", sub = "byte")
if (
  length(lines) < 2 ||
  str_detect(str_to_lower(lines[1]), "<html|<!doctype|wafalert")
) {
  stop("Attendance source did not return a tab-delimited data file.")
}

readr::read_tsv(
  I(paste(lines, collapse = "\n")),
  col_types = readr::cols(.default = readr::col_character()),
  na = c("", "NA"),
  show_col_types = FALSE
)
}

read_chronic_absenteeism_file <- function(file_name, url, closure_year) {
  source <- resolve_attendance_source(file_name, url)
  old_timeout <- getOption("timeout")
  options(timeout = 20)
  on.exit(options(timeout = old_timeout), add = TRUE)

  tryCatch(
    {
      chronic_data <- read_tsv_source(source) |>
        clean_names() |>
        standardize_chronic_absenteeism_names()

      required_chronic_cols <- c(
        "academic_year",
        "aggregate_level",
        "county_code",
        "county_name",
        "charter",
        "dass",
        "reporting_category",
        "chronic_absenteeism_eligible_cumulative_enrollment",
        "chronic_absenteeism_count"
      )
    }
  )
}

```



```

missing_chronic_cols <- setdiff(required_chronic_cols, names(chronic_data))
if (length(missing_chronic_cols) > 0) {
  stop(
    "Chronic absenteeism file did not contain expected columns: ",
    str_c(missing_chronic_cols, collapse = ", ")
  )
}

chronic_data |>
  mutate(
    closure_year = closure_year,
    source_url = source,
    chronic_absenteeism_read_status = "ok"
  )
},
error = function(e) {
  empty_chronic_absenteeism(
    closure_year = closure_year,
    source = source,
    status = str_c("read failed: ", conditionMessage(e))
  )
}
)
}

chronic_absenteeism_sources <- tibble(
  academic_year = analysis_year_labels,
  closure_year = analysis_years,
  file_name = c(
    "chronicabsenteeism18.txt",
    "chronicabsenteeism19.txt"
  ),
  url = c(
    "https://www3.cde.ca.gov/demo-downloads/attendance/chronicabsenteeism18.txt",
    "https://www3.cde.ca.gov/demo-downloads/attendance/chronicabsenteeism19.txt"
  )
)

chronic_absenteeism_raw <- chronic_absenteeism_sources |>
  mutate(data = pmap(list(file_name, url, closure_year), read_chronic_absenteeism_file)) |>
  select(data) |>
  unnest(cols = data)

```

```
chronic_absenteeism_raw |>
  distinct(closure_year, source_url, chronic_absenteeism_read_status)
```

```
# A tibble: 2 x 3
  closure_year source_url      chronic_absenteeism_read_status
    <dbl> <chr>                <chr>
1    2018 https://www3.cde.ca.gov/demo-downloads/at~ ok
2    2019 https://www3.cde.ca.gov/demo-downloads/at~ ok
# i abbreviated name: 1: chronic_absenteeism_read_status
```

Chronic absenteeism data quality checks

```
chronic_absenteeism_county_year <- chronic_absenteeism_raw |>
  filter(
    chronic_absenteeism_read_status == "ok",
    aggregate_level == "C",
    reporting_category == "TA",
    str_to_lower(charter) == "all",
    str_to_lower(dass) == "all"
  ) |>
  mutate(
    county_code = pad_code(county_code, 2),
    county_name = str_squish(str_to_title(county_name)),
    across(all_of(chronic_absenteeism_numeric_vars), to_number),
    chronic_absenteeism_rate_reported = chronic_absenteeism_rate_reported / 100,
    chronic_absenteeism_rate = safe_divide(
      chronic_absenteeism_count,
      chronic_absenteeism_eligible_cumulative_enrollment
    ),
    chronic_absenteeism_rate_reported_difference = chronic_absenteeism_rate -
      chronic_absenteeism_rate_reported
  )

chronic_absenteeism_county_year |>
  summarise(
    rows = n(),
    county_years = n_distinct(str_c(county_name, closure_year)),
    counties = n_distinct(county_name),
    years = n_distinct(closure_year),
```

```

suppressed_or_missing_counties = sum(
  is.na(chronic_absenteeism_eligible_cumulative_enrollment) |
  is.na(chronic_absenteeism_count)
)
)

```

```

# A tibble: 1 x 5
  rows county_years counties years suppressed_or_missing_counties
<int>      <int>      <int> <int>                      <int>
1   116        116        58    2                        0

```

```

chronic_absenteeism_county_year |>
  count(county_name, closure_year, sort = TRUE) |>
  filter(n != 1) |>
  slice_head(n = 20)

```

```

# A tibble: 0 x 3
# i 3 variables: county_name <chr>, closure_year <dbl>, n <int>

```

```

chronic_absenteeism_county_year |>
  summarise(
    median_abs_rate_difference = median(
      abs(chronic_absenteeism_rate_reported_difference),
      na.rm = TRUE
    ),
    max_abs_rate_difference = max(
      abs(chronic_absenteeism_rate_reported_difference),
      na.rm = TRUE
    )
  )

```

```

# A tibble: 1 x 2
  median_abs_rate_difference max_abs_rate_difference
                        <dbl>                      <dbl>
1                0.000271                0.000495

```

```

county_chronic_absenteeism <- chronic_absenteeism_county_year |>
  group_by(county_name) |>
  summarise(
    chronic_years_observed = n_distinct(closure_year),

```

```

    chronic_absenteeism_eligible_cumulative_enrollment = sum(
      chronic_absenteeism_eligible_cumulative_enrollment,
      na.rm = TRUE
    ),
    chronic_absenteeism_count = sum(chronic_absenteeism_count, na.rm = TRUE),
    pooled_chronic_absenteeism_rate = safe_divide(
      chronic_absenteeism_count,
      chronic_absenteeism_eligible_cumulative_enrollment
    ),
    chronic_absenteeism_rate_2017_18 = safe_divide(
      sum(chronic_absenteeism_count[closure_year == 2018], na.rm = TRUE),
      sum(
        chronic_absenteeism_eligible_cumulative_enrollment[closure_year == 2018],
        na.rm = TRUE
      )
    ),
    chronic_absenteeism_rate_2018_19 = safe_divide(
      sum(chronic_absenteeism_count[closure_year == 2019], na.rm = TRUE),
      sum(
        chronic_absenteeism_eligible_cumulative_enrollment[closure_year == 2019],
        na.rm = TRUE
      )
    ),
    .groups = "drop"
  ) |>
  filter(chronic_absenteeism_eligible_cumulative_enrollment > 0)

if (nrow(county_chronic_absenteeism) == 0) {
  tibble(
    chronic_absenteeism_status = "No county chronic absenteeism rows are available. Add local"
  )
} else {
  county_chronic_absenteeism |>
    arrange(desc(pooled_chronic_absenteeism_rate)) |>
    select(
      county_name,
      chronic_years_observed,
      chronic_absenteeism_eligible_cumulative_enrollment,
      chronic_absenteeism_count,
      pooled_chronic_absenteeism_rate,
      chronic_absenteeism_rate_2017_18,
      chronic_absenteeism_rate_2018_19
    )
}

```

```

) |>
  slice_head(n = 15)
}

```

```
# A tibble: 15 x 7
```

	county_name <chr>	chronic_years_observed <int>	chronic_absenteeism_eligible_cumulativ~1 <dbl>
1	Inyo	2	11279
2	Nevada	2	25122
3	Plumas	2	2749
4	Lake	2	20109
5	Trinity	2	3416
6	Mono	2	4342
7	Mendocino	2	27034
8	Mariposa	2	3966
9	Modoc	2	2950
10	San Francisco	2	131884
11	Del Norte	2	8862
12	Alpine	2	169
13	Siskiyou	2	12551
14	Butte	2	65296
15	Humboldt	2	37608

```
# i abbreviated name: 1: chronic_absenteeism_eligible_cumulative_enrollment
```

```
# i 4 more variables: chronic_absenteeism_count <dbl>,
```

```
#   pooled_chronic_absenteeism_rate <dbl>,
```

```
#   chronic_absenteeism_rate_2017_18 <dbl>,
```

```
#   chronic_absenteeism_rate_2018_19 <dbl>
```

```

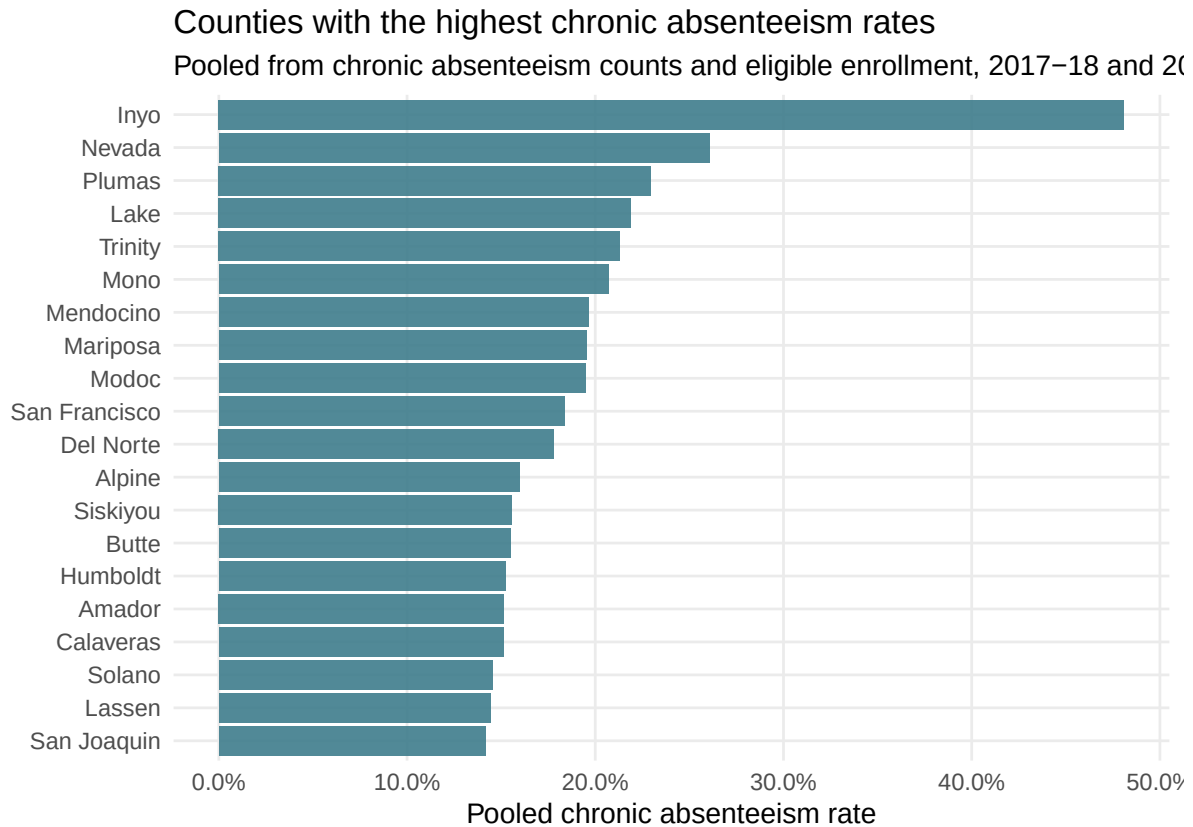
if (nrow(county_chronic_absenteeism) > 0) {
  county_chronic_absenteeism |>
    slice_max(pooled_chronic_absenteeism_rate, n = 20, with_ties = FALSE) |>
    ggplot(
      aes(
        x = pooled_chronic_absenteeism_rate,
        y = fct_reorder(county_name, pooled_chronic_absenteeism_rate)
      )
    ) +
    geom_col(fill = "#3F7D8C", alpha = 0.9) +
    scale_x_continuous(labels = label_percent(accuracy = 0.1)) +
    labs(
      title = "Counties with the highest chronic absenteeism rates",

```

```

    subtitle = "Pooled from chronic absenteeism counts and eligible enrollment, 2017-18 and 2018-19",
    x = "Pooled chronic absenteeism rate",
    y = NULL
  ) +
  theme(panel.grid.minor = element_blank())
}

```



Step 3 takeaway

The chronic absenteeism files provide the appropriate county-level outcome for the question about chronic absenteeism rates. The main rate used here is pooled from chronic absenteeism counts and eligible cumulative enrollment across 2017-18 and 2018-19, rather than averaging county-year percentages.

Step 4: Do higher wildfire-burden counties also have higher chronic absenteeism rates?

```
county_wildfire_top_cutoff <- county_window_burden |>
  filter(wildfire_student_days > 0) |>
  summarise(cutoff = quantile(wildfire_student_days, 0.75, na.rm = TRUE)) |>
  pull(cutoff)

if (length(county_wildfire_top_cutoff) == 0 || is.na(county_wildfire_top_cutoff)) {
  county_wildfire_top_cutoff <- Inf
}

county_wildfire_chronic <- county_window_burden |>
  full_join(
    county_chronic_absenteeism,
    by = c("county" = "county_name")
  ) |>
  mutate(
    wildfire_student_days = replace_na(wildfire_student_days, 0),
    wildfire_days = replace_na(wildfire_days, 0),
    total_enrollment_school_years = replace_na(total_enrollment_school_years, 0),
    wildfire_student_days_per_1k_enrolled = safe_divide(
      wildfire_student_days * 1000,
      total_enrollment_school_years
    ),
    county_wildfire_burden_group = case_when(
      wildfire_student_days <= 0 ~ "No wildfire",
      wildfire_student_days >= county_wildfire_top_cutoff ~ "Affected: top 25%",
      TRUE ~ "Affected: lower 75%"
    ),
    county_wildfire_burden_group = factor(
      county_wildfire_burden_group,
      levels = c("No wildfire", "Affected: lower 75%", "Affected: top 25%")
    )
  )

county_wildfire_chronic |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    wildfire_days,
    wildfire_student_days,
```

```
wildfire_student_days_per_1k_enrolled,
chronic_years_observed,
chronic_absenteeism_eligible_cumulative_enrollment,
pooled_chronic_absenteeism_rate
) |>
slice_head(n = 15)
```

```
# A tibble: 15 x 7
```

	county	wildfire_days	wildfire_student_days	wildfire_student_days_per~1
	<chr>	<dbl>	<dbl>	<dbl>
1	Ventura	1540	1052437	3849.
2	Sonoma	1805	755475	5389.
3	Los Angeles	879	635624	215.
4	Contra Costa	596	364434	1024.
5	Riverside	371	362921	423.
6	Butte	928	324612	5103.
7	Santa Barbara	570	271895	1954.
8	Napa	447	219721	5410.
9	Sacramento	305	209115	425.
10	Solano	253	163051	1288.
11	Alameda	178	129436	284.
12	San Diego	168	111207	110.
13	Lake	344	88748	4624.
14	Marin	222	86526	1288.
15	Orange	107	78093	81.0

```
# i abbreviated name: 1: wildfire_student_days_per_1k_enrolled
# i 3 more variables: chronic_years_observed <int>,
#   chronic_absenteeism_eligible_cumulative_enrollment <dbl>,
#   pooled_chronic_absenteeism_rate <dbl>
```

```
county_wildfire_chronic |>
  filter(!is.na(pooled_chronic_absenteeism_rate)) |>
  arrange(desc(pooled_chronic_absenteeism_rate)) |>
  select(
    county,
    pooled_chronic_absenteeism_rate,
    wildfire_days,
    wildfire_student_days,
    wildfire_student_days_per_1k_enrolled,
    county_wildfire_burden_group
  ) |>
  slice_head(n = 15)
```



```
# A tibble: 15 x 6
```

	county <chr>	pooled_chronic_absenteeism_rate <dbl>	wildfire_days <dbl>	wildfire_student_days <dbl>
1	Inyo	0.481	0	0
2	Nevada	0.261	29	9478
3	Plumas	0.230	0	0
4	Lake	0.219	344	88748
5	Trinity	0.213	36	6273
6	Mono	0.207	4	392
7	Mendocino	0.196	181	53199
8	Mariposa	0.195	0	0
9	Modoc	0.195	0	0
10	San Francisco	0.184	8	2275
11	Del Norte	0.178	0	0
12	Alpine	0.160	0	0
13	Siskiyou	0.156	2	34
14	Butte	0.155	928	324612
15	Humboldt	0.152	21	3003

```
# i abbreviated name: 1: pooled_chronic_absenteeism_rate  
# i 2 more variables: wildfire_student_days_per_1k_enrolled <dbl>,  
#   county_wildfire_burden_group <fct>
```

```
county_wildfire_chronic_group_summary <- county_wildfire_chronic |>  
  filter(!is.na(pooled_chronic_absenteeism_rate)) |>  
  group_by(county_wildfire_burden_group) |>  
  summarise(  
    counties = n(),  
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),  
    chronic_absenteeism_eligible_cumulative_enrollment = sum(  
      chronic_absenteeism_eligible_cumulative_enrollment,  
      na.rm = TRUE  
    ),  
    chronic_absenteeism_count = sum(chronic_absenteeism_count, na.rm = TRUE),  
    median_county_chronic_absenteeism_rate = median(  
      pooled_chronic_absenteeism_rate,  
      na.rm = TRUE  
    ),  
    pooled_chronic_absenteeism_rate = safe_divide(  
      chronic_absenteeism_count,  
      chronic_absenteeism_eligible_cumulative_enrollment  
    ),  
    .groups = "drop"
```

```
)

county_wildfire_chronic_group_summary

# A tibble: 3 x 7
  county_wildfire_burden~1 counties wildfire_student_days chronic_absenteeism~2
  <fct>                  <int>          <dbl>          <dbl>
1 No wildfire            22            0            2511155
2 Affected: lower 75%    27          1095674          4783464
3 Affected: top 25%      9          4196234          5426528
# i abbreviated names: 1: county_wildfire_burden_group,
#   2: chronic_absenteeism_eligible_cumulative_enrollment
# i 3 more variables: chronic_absenteeism_count <dbl>,
#   median_county_chronic_absenteeism_rate <dbl>,
#   pooled_chronic_absenteeism_rate <dbl>
```

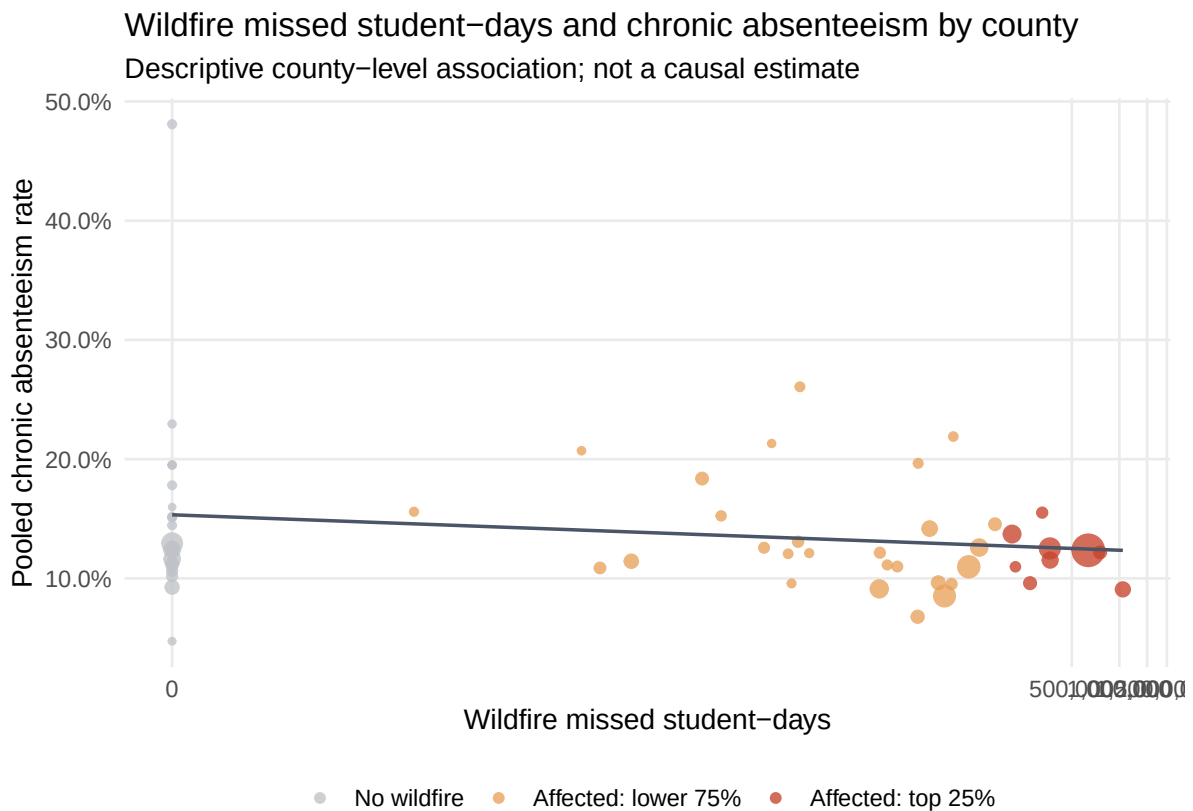
```
county_wildfire_chronic_plot_data <- county_wildfire_chronic |>
  filter(!is.na(pooled_chronic_absenteeism_rate))

if (nrow(county_wildfire_chronic_plot_data) > 1) {
  ggplot(
    county_wildfire_chronic_plot_data,
    aes(x = wildfire_student_days, y = pooled_chronic_absenteeism_rate)
  ) +
    geom_point(
      aes(
        size = chronic_absenteeism_eligible_cumulative_enrollment,
        color = county_wildfire_burden_group
      ),
      alpha = 0.78
    ) +
    geom_smooth(method = "lm", se = FALSE, color = "#4A5568", linewidth = 0.7) +
    scale_x_continuous(
      trans = pseudo_log_trans(base = 10),
      labels = label_comma()
    ) +
    scale_y_continuous(labels = label_percent(accuracy = 0.1)) +
    scale_size_continuous(labels = label_comma(), guide = "none") +
    scale_color_manual(values = burden_palette, name = NULL) +
    labs(
      title = "Wildfire missed student-days and chronic absenteeism by county",
```

```

    subtitle = "Descriptive county-level association; not a causal estimate",
    x = "Wildfire missed student-days",
    y = "Pooled chronic absenteeism rate"
  ) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  )
} else {
  tibble(chronic_absenteeism_status = "Not enough county chronic absenteeism data to draw the")
}

```



```

county_wildfire_chronic_sensitivity <- county_wildfire_chronic |>
  filter(!is.na(pooled_chronic_absenteeism_rate)) |>
  select(
    county,
    wildfire_student_days,

```

```

wildfire_student_days_per_1k_enrolled,
pooled_chronic_absenteeism_rate,
county_wildfire_burden_group
) |>
arrange(desc(wildfire_student_days_per_1k_enrolled))

county_wildfire_chronic_sensitivity |>
  slice_head(n = 15)

```

```
# A tibble: 15 x 5
```

	county	wildfire_student_days	wildfire_student_day~1	pooled_chronic_absen~2
	<chr>	<dbl>	<dbl>	<dbl>
1	Napa	219721	5410.	0.110
2	Sonoma	755475	5389.	0.122
3	Butte	324612	5103.	0.155
4	Lake	88748	4624.	0.219
5	Ventura	1052437	3849.	0.0907
6	Mendocino	53199	2021.	0.196
7	Trinity	6273	1972.	0.213
8	Santa Ba~	271895	1954.	0.0960
9	Marin	86526	1288.	0.0952
10	Solano	163051	1288.	0.145
11	Yuba	33869	1170.	0.111
12	Contra C~	364434	1024.	0.115
13	Glenn	10848	964.	0.121
14	Colusa	8391	902.	0.0957
15	Sutter	39253	809.	0.110

```

# i abbreviated names: 1: wildfire_student_days_per_1k_enrolled,
#   2: pooled_chronic_absenteeism_rate
# i 1 more variable: county_wildfire_burden_group <fct>

```

```

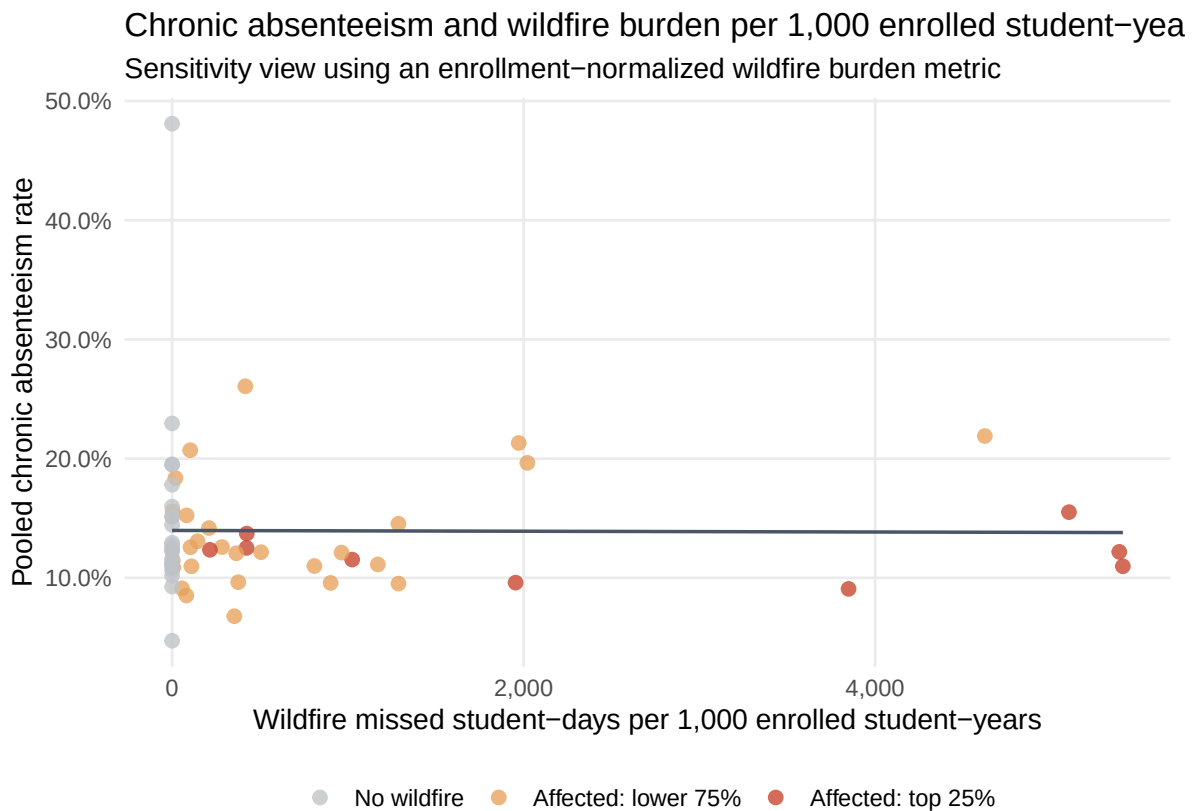
if (nrow(county_wildfire_chronic_sensitivity) > 1) {
  ggplot(
    county_wildfire_chronic_sensitivity,
    aes(
      x = wildfire_student_days_per_1k_enrolled,
      y = pooled_chronic_absenteeism_rate
    )
  ) +
  geom_point(aes(color = county_wildfire_burden_group), size = 2.4, alpha = 0.78) +
  geom_smooth(method = "lm", se = FALSE, color = "#4A5568", linewidth = 0.7) +

```

```

scale_x_continuous(labels = label_comma()) +
scale_y_continuous(labels = label_percent(accuracy = 0.1)) +
scale_color_manual(values = burden_palette, name = NULL) +
labs(
  title = "Chronic absenteeism and wildfire burden per 1,000 enrolled student-years",
  subtitle = "Sensitivity view using an enrollment-normalized wildfire burden metric",
  x = "Wildfire missed student-days per 1,000 enrolled student-years",
  y = "Pooled chronic absenteeism rate"
) +
theme(
  legend.position = "bottom",
  panel.grid.minor = element_blank()
)
}

```



Step 4 takeaway

This county-level comparison directly addresses whether the counties with the most wildfire disruption also have higher chronic absenteeism rates. It remains descriptive: counties differ in enrollment size, baseline attendance risk, geography, socioeconomic context, and reporting conditions, so the plots should be read as pattern-finding rather than evidence that wildfire closures caused chronic absenteeism.

Step 5: What do absence-day patterns look like in affected schools?

Load CDE Absenteeism by Reason data

CDE file source: <https://www.cde.ca.gov/ds/ad/filesabr.asp>

```
absenteeism_col_names <- c(
  "academic_year",
  "aggregate_level",
  "county_code",
  "district_code",
  "school_code",
  "county_name",
  "district_name",
  "school_name",
  "charter",
  "dass",
  "reporting_category",
  "eligible_cumulative_enrollment",
  "count_students_one_or_more_absences",
  "average_days_absent",
  "total_days_absent",
  "excused_absences_percent",
  "unexcused_absences_percent",
  "out_of_school_suspension_absences_percent",
  "incomplete_independent_study_absences_percent",
  "excused_absences_count",
  "unexcused_absences_count",
  "out_of_school_suspension_absences_count",
  "incomplete_independent_study_absences_count"
)

absenteeism_numeric_vars <- c(
  "eligible_cumulative_enrollment",
```

```

"count_students_one_or_more_absences",
"average_days_absent",
"total_days_absent",
"excused_absences_percent",
"unexcused_absences_percent",
"out_of_school_suspension_absences_percent",
"incomplete_independent_study_absences_percent",
"excused_absences_count",
"unexcused_absences_count",
"out_of_school_suspension_absences_count",
"incomplete_independent_study_absences_count"
)

absence_reason_count_vars <- c(
  "excused_absences_count",
  "unexcused_absences_count",
  "out_of_school_suspension_absences_count",
  "incomplete_independent_study_absences_count"
)

read_absenteeism_file <- function(file_name, url, closure_year) {
  source <- resolve_attendance_source(file_name, url)

  readr::read_tsv(
    source,
    skip = 1,
    col_names = absenteeism_col_names,
    col_types = readr::cols(.default = readr::col_character()),
    na = c("", "NA")
  ) |>
  clean_names() |>
  mutate(
    closure_year = closure_year,
    source_url = source
  )
}

absenteeism_sources <- tibble(
  academic_year = analysis_year_labels,
  closure_year = analysis_years,
  file_name = c(
    "absenteeismreason18.txt",

```

```

      "absenteeismreason19.txt"
    ),
    url = c(
      "https://www3.cde.ca.gov/demo-downloads/attendance/absenteeismreason18.txt",
      "https://www3.cde.ca.gov/demo-downloads/attendance/absenteeismreason19.txt"
    )
  )

absenteeism_raw <- absenteeism_sources |>
  mutate(data = pmap(list(file_name, url, closure_year), read_absenteeism_file)) |>
  select(data) |>
  unnest(cols = data)

absenteeism_schools <- absenteeism_raw |>
  filter(
    aggregate_level == "S",
    reporting_category == "TA"
  ) |>
  mutate(
    suppressed_numeric_fields = rowSums(across(all_of(absenteeism_numeric_vars), ~ .x == "*")),
    county_code_clean = pad_code(county_code, 2),
    district_code_clean = pad_code(district_code, 5),
    school_code_clean = pad_code(school_code, 7),
    cds_code = if_else(
      !is.na(county_code_clean) & !is.na(district_code_clean) & !is.na(school_code_clean),
      str_c(county_code_clean, district_code_clean, school_code_clean),
      NA_character_
    ),
    across(all_of(absenteeism_numeric_vars), to_number),
    county_name = str_squish(str_to_title(county_name)),
    district_name = str_squish(district_name),
    school_name = str_squish(school_name),
    absence_reason_count_sum = rowSums(pick(all_of(absence_reason_count_vars))), na.rm = TRUE,
    absence_reason_count_difference = total_days_absent - absence_reason_count_sum,
    share_students_with_absence = safe_divide(
      count_students_one_or_more_absences,
      eligible_cumulative_enrollment
    ),
    avg_days_absent_among_absent = average_days_absent,
    avg_days_absent_per_eligible = safe_divide(
      total_days_absent,
      eligible_cumulative_enrollment
    )
  )

```



```
),
excused_absence_share = safe_divide(excused_absences_count, total_days_absent),
unexcused_absence_share = safe_divide(unexcused_absences_count, total_days_absent),
suspension_absence_share = safe_divide(out_of_school_suspension_absences_count, total_days_absent),
incomplete_independent_study_absence_share = safe_divide(
    incomplete_independent_study_absences_count,
    total_days_absent
)
)

glimpse(absenteeism_schools)
```

Rows: 20,227

Columns: 39

```
$ academic_year
18", "2~
```

```
$ aggregate_level
```

```
$ county_code
```

```
$ district_code
```

```
$ school_code
```

```
$ county_name
```

```
$ district_name
```

```
$ school_name
```

\$ charter

\$ dass

```
$ reporting_category
```

```
$ eligible_cumulative_enrollment
```

```
$ count_students_one_or_more_absences
```

```
$ average_days_absent
```

```
$ total_days_absent
```

\$ excused_absences_percent

\$ unexcused_absences_percent

```
$ out_of_school_suspension_absences_percent
```

```
$ incomplete_independent_study_absences_percent
```

```
$ excused_absences_count
```

```
$ unexcused_absences_count
```

```
$ out_of_school_suspension_absences_count
```

```
$ incomplete_independent_study_absences_count
```

```
$ closure_year
```

```
$ source_url
```

\$ suppressed_numeric_fields

```
$ county_code_clean
```

```

$ district_code_clean      <chr> "10017", "10017", "10017~
$ school_code_clean       <chr> "0130419", "0130401", "0~
$ cds_code                <chr> "01100170130419", "01100~
$ absence_reason_count_sum <dbl> 9675.0, 526.0, 2678.0, 1~
$ absence_reason_count_difference <dbl> 0, 0, 0, 0, 0, 0, 0, 0, ~
$ share_students_with_absence <dbl> 0.9440789, 0.6566265, 0.~
$ avg_days_absent_among_absent <dbl> 33.7, 4.8, 10.7, 1.9, 9.~
$ avg_days_absent_per_eligible <dbl> 31.8256579, 3.1686747, 9~
$ excused_absence_share    <dbl> 0.1352972, 0.6140684, 0.~
$ unexcused_absence_share  <dbl> 0.3120413, 0.3669202, 0.~
$ suspension_absence_share <dbl> 0.0043410853, 0.01901140~
$ incomplete_independent_study_absence_share <dbl> 0.548320413, 0.000000000~

```

Absenteeism data quality checks

```

absenteeism_schools |>
  summarise(
    rows = n(),
    school_years = n_distinct(str_c(cds_code, closure_year)),
    schools = n_distinct(cds_code),
    rows_with_suppressed_values = sum(suppressed_numeric_fields > 0, na.rm = TRUE),
    rows_missing_cds_code = sum(is.na(cds_code))
  )

```

```

# A tibble: 1 x 5
  rows school_years schools rows_with_suppressed_values rows_missing_cds_code
  <int>      <int>   <int>                <int>                <int>
1 20227      20227  10336                  309                  0

```

```

absenteeism_schools |>
  count(cds_code, closure_year, sort = TRUE) |>
  filter(n > 1) |>
  slice_head(n = 20)

```

```

# A tibble: 0 x 3
# i 3 variables: cds_code <chr>, closure_year <dbl>, n <int>

```

```
absenteeism_schools |>
  summarise(
    across(
      all_of(absenteeism_numeric_vars),
      ~ mean(is.na(.x)),
      .names = "missing_share_{.col}"
    )
  ) |>
  pivot_longer(everything(), names_to = "field", values_to = "missing_share") |>
  arrange(desc(missing_share)) |>
  slice_head(n = 20)
```

```
# A tibble: 12 x 2
```

	field	missing_share
	<chr>	<dbl>
1	missing_share_eligible_cumulative_enrollment	0.0153
2	missing_share_count_students_one_or_more_absences	0.0153
3	missing_share_average_days_absent	0.0153
4	missing_share_total_days_absent	0.0153
5	missing_share_excused_absences_percent	0.0153
6	missing_share_unexcused_absences_percent	0.0153
7	missing_share_out_of_school_suspension_absences_percent	0.0153
8	missing_share_incomplete_independent_study_absences_percent	0.0153
9	missing_share_excused_absences_count	0.0153
10	missing_share_unexcused_absences_count	0.0153
11	missing_share_out_of_school_suspension_absences_count	0.0153
12	missing_share_incomplete_independent_study_absences_count	0.0153

```
absenteeism_schools |>
  mutate(abs_reason_abs_diff = abs(absence_reason_count_difference)) |>
  summarise(
    rows = n(),
    median_abs_difference = median(abs_reason_abs_diff, na.rm = TRUE),
    p95_abs_difference = quantile(abs_reason_abs_diff, 0.95, na.rm = TRUE),
    max_abs_difference = max(abs_reason_abs_diff, na.rm = TRUE)
  )
```

```
# A tibble: 1 x 4
```

	rows	median_abs_difference	p95_abs_difference	max_abs_difference
	<int>	<dbl>	<dbl>	<dbl>
1	20227	0	0	0.200

Join closure school-years to absenteeism school-years

```
closure_school_year <- closure_window |>
  group_by(cds_code, closure_year) |>
  summarise(
    academic_year = first(academic_year),
    county = first(county),
    district_name = first(district_name),
    school_site = first(school_site),
    total_enrollment = max(total_enrollment, na.rm = TRUE),
    days_wildfire = sum(days_wildfire, na.rm = TRUE),
    days_total = sum(days_total, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    wildfire_share_of_total = safe_divide(days_wildfire, days_total),
    any_wildfire = any(days_wildfire > 0, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(total_enrollment = if_else(is.infinite(total_enrollment), NA_real_, total_enrollment))

closure_absenteeism_joined <- closure_school_year |>
  left_join(
    absenteeism_schools |>
      select(
        cds_code,
        closure_year,
        absenteeism_academic_year = academic_year,
        county_name_absenteeism = county_name,
        district_name_absenteeism = district_name,
        school_name_absenteeism = school_name,
        charter,
        dass,
        eligible_cumulative_enrollment,
        count_students_one_or_more_absences,
        average_days_absent,
        total_days_absent,
        all_of(absence_reason_count_vars),
        share_students_with_absence,
        avg_days_absent_among_absent,
        avg_days_absent_per_eligible,
        excused_absence_share,
        unexcused_absence_share,
        suspension_absence_share,
```

```

      incomplete_independent_study_absence_share,
      suppressed_numeric_fields
    ),
    by = c("cds_code", "closure_year")
  )

affected_cutoff <- closure_absenteeism_joined |>
  filter(wildfire_student_days > 0, !is.na(avg_days_absent_per_eligible)) |>
  summarise(cutoff = quantile(wildfire_student_days, 0.75, na.rm = TRUE)) |>
  pull(cutoff)

if (length(affected_cutoff) == 0 || is.na(affected_cutoff)) {
  affected_cutoff <- Inf
}

closure_absenteeism_joined <- closure_absenteeism_joined |>
  mutate(
    burden_group = case_when(
      wildfire_student_days <= 0 ~ "No wildfire",
      wildfire_student_days >= affected_cutoff ~ "Affected: top 25%",
      TRUE ~ "Affected: lower 75%"
    ),
    burden_group = factor(
      burden_group,
      levels = c("No wildfire", "Affected: lower 75%", "Affected: top 25%")
    )
  )

join_coverage <- tibble(
  metric = c(
    "closure_school_year_rows",
    "closure_school_year_rows_matched_to_absenteeism",
    "closure_school_year_match_rate",
    "unmatched_closure_school_year_rows",
    "absenteeism_school_year_rows_not_in_closure_data"
  ),
  value = c(
    nrow(closure_school_year),
    sum(!is.na(closure_absenteeism_joined$eligible_cumulative_enrollment)),
    safe_divide(
      sum(!is.na(closure_absenteeism_joined$eligible_cumulative_enrollment)),
      nrow(closure_school_year)
    )
  )

```

```

    ),
    nrow(anti_join(closure_school_year, absenteeism_schools, by = c("cds_code", "closure_year")),
    nrow(anti_join(absenteeism_schools, closure_school_year, by = c("cds_code", "closure_year"))
  )
)

join_coverage

```

```

# A tibble: 5 x 2
  metric                                value
  <chr>                                <dbl>
1 closure_school_year_rows             23258
2 closure_school_year_rows_matched_to_absenteeism 19867
3 closure_school_year_match_rate        0.854
4 unmatched_closure_school_year_rows    3158
5 absenteeism_school_year_rows_not_in_closure_data 127

```

The join covers most closure school-year rows, but not all of them. Unmatched closure rows remain in the closure data but do not contribute to absenteeism summaries, and absenteeism rows not found in the closure data are excluded from the closure-centered join. The unmatched examples below suggest that some non-matches are likely institutional or nonstandard records, but this notebook does not assume every unmatched row has the same reason.

```

anti_join(closure_school_year, absenteeism_schools, by = c("cds_code", "closure_year")) |>
  select(cds_code, closure_year, county, district_name, school_site, days_wildfire, days_total) |>
  arrange(desc(days_wildfire), county, district_name, school_site) |>
  slice_head(n = 20)

```

```

# A tibble: 20 x 7
  cds_code      closure_year county  district_name school_site days_wildfire
  <chr>          <int> <chr>    <chr>          <chr>          <dbl>
1 49708620138560    2018 Sonoma    Petaluma Joi~ RISE Acade~      5
2 19101990128025    2018 Los Ange~ Los Angeles ~ Lashon Aca~      3
3 07616550110528    2019 Contra C~ Brentwood Un~ Adams (J. ~      1
4 07616556003644    2019 Contra C~ Brentwood Un~ Brentwood ~      1
5 07616556003651    2019 Contra C~ Brentwood Un~ Edna Hill ~      1
6 07616556070981    2019 Contra C~ Brentwood Un~ Garin Elem~      1
7 07616556117410    2019 Contra C~ Brentwood Un~ Loma Vista~      1
8 07616550108019    2019 Contra C~ Brentwood Un~ Marsh Cree~      1
9 07616550127464    2019 Contra C~ Brentwood Un~ Mary Casey~      1
10 07616550102376    2019 Contra C~ Brentwood Un~ Pioneer El~      1

```

```

11 07616556119754      2019 Contra C~ Brentwood Un~ R. Paul Kr~      1
12 07616556109730      2019 Contra C~ Brentwood Un~ Ron Nunn E~      1
13 07616556112718      2019 Contra C~ Brentwood Un~ William B.~      1
14 01100176106751      2018 Alameda   Alameda Coun~ Alameda Co~      0
15 01100176106751      2019 Alameda   Alameda Coun~ Alameda Co~      0
16 01100170137448      2018 Alameda   Alameda Coun~ Aurum Prep~      0
17 01100170129403      2018 Alameda   Alameda Coun~ Epic Chart~      0
18 01100170129403      2019 Alameda   Alameda Coun~ Epic Chart~      0
19 01100170138867      2018 Alameda   Alameda Coun~ Hayward Co~      0
20 01100170138867      2019 Alameda   Alameda Coun~ Hayward Co~      0
# i 1 more variable: days_total <dbl>

```

```

anti_join(absenteeism_schools, closure_school_year, by = c("cds_code", "closure_year")) |>
  select(cds_code, closure_year, county_name, district_name, school_name, eligible_cumulative) |>
  arrange(county_name, district_name, school_name) |>
  slice_head(n = 20)

```

```

# A tibble: 20 x 6
  cds_code      closure_year county_name district_name      school_name
  <chr>          <dbl> <chr>      <chr>          <chr>
1 01750930000000 2019 Alameda   Dublin Unified District 0~
2 01751010000000 2018 Alameda   Pleasanton Unified District 0~
3 01751010000000 2019 Alameda   Pleasanton Unified District 0~
4 07616550000000 2018 Contra Costa Brentwood Union Element~ District 0~
5 071007400000001 2019 Contra Costa Contra Costa County Off~ Nonpublic,~
6 076170500000001 2018 Contra Costa Knightsen Elementary Nonpublic,~
7 07617540000000 2018 Contra Costa Mt. Diablo Unified District 0~
8 07617540000000 2019 Contra Costa Mt. Diablo Unified District 0~
9 096190300000001 2018 El Dorado   Lake Tahoe Unified Nonpublic,~
10 106212500000001 2019 Fresno      Coalinga-Huron Unified Nonpublic,~
11 106233100000001 2019 Fresno      Orange Center Nonpublic,~
12 106236400000001 2018 Fresno      Parlier Unified Nonpublic,~
13 12755150000000 2018 Humboldt    Eureka City Schools District 0~
14 12755150000000 2019 Humboldt    Eureka City Schools District 0~
15 15635290000000 2019 Kern        Kern High District 0~
16 156352900000001 2019 Kern        Kern High Nonpublic,~
17 156362800000001 2019 Kern        Maricopa Unified Nonpublic,~
18 156336200000001 2019 Kern        Panama-Buena Vista Union Nonpublic,~
19 16639580000000 2018 Kings       Kit Carson Union Elemen~ District 0~
20 19642120000000 2018 Los Angeles ABC Unified District 0~
# i 1 more variable: eligible_cumulative_enrollment <dbl>

```

Absenteeism by wildfire burden group

```
burden_group_summary <- closure_absenteeism_joined |>
  filter(!is.na(avg_days_absent_per_eligible)) |>
  group_by(burden_group) |>
  summarise(
    school_year_rows = n(),
    schools = n_distinct(cds_code),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    eligible_cumulative_enrollment = sum(eligible_cumulative_enrollment, na.rm = TRUE),
    count_students_one_or_more_absences = sum(count_students_one_or_more_absences, na.rm = TRUE),
    total_days_absent = sum(total_days_absent, na.rm = TRUE),
    share_students_with_absence = safe_divide(
      count_students_one_or_more_absences,
      eligible_cumulative_enrollment
    ),
    avg_days_absent_per_eligible = safe_divide(total_days_absent, eligible_cumulative_enrollment),
    avg_days_absent_among_absent = safe_divide(
      total_days_absent,
      count_students_one_or_more_absences
    ),
    .groups = "drop"
  )

burden_group_summary
```

```
# A tibble: 3 x 11
  burden_group      school_year_rows schools wildfire_days wildfire_student_days
  <fct>              <int>    <int>         <dbl>             <dbl>
1 No wildfire        16548     9457             0                 0
2 Affected: lower ~   2489     2133            4747            1631238
3 Affected: top 25%    830      780            4888            3650974
# i 6 more variables: eligible_cumulative_enrollment <dbl>,
#   count_students_one_or_more_absences <dbl>, total_days_absent <dbl>,
#   share_students_with_absence <dbl>, avg_days_absent_per_eligible <dbl>,
#   avg_days_absent_among_absent <dbl>
```

```
top_10_cutoff <- closure_absenteeism_joined |>
  filter(wildfire_student_days > 0, !is.na(avg_days_absent_per_eligible)) |>
  summarise(cutoff = quantile(wildfire_student_days, 0.90, na.rm = TRUE)) |>
```



```

pull(cutoff)

if (length(top_10_cutoff) == 0 || is.na(top_10_cutoff)) {
  top_10_cutoff <- Inf
}

make_burden_sensitivity <- function(cutoff, top_label, definition_label) {
  closure_absenteeism_joined |>
    filter(!is.na(avg_days_absent_per_eligible)) |>
    mutate(
      sensitivity_group = case_when(
        wildfire_student_days <= 0 ~ "No wildfire",
        wildfire_student_days >= cutoff ~ top_label,
        TRUE ~ "Affected: below cutoff"
      )
    ) |>
    group_by(sensitivity_group) |>
    summarise(
      school_year_rows = n(),
      schools = n_distinct(cds_code),
      wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
      eligible_cumulative_enrollment = sum(eligible_cumulative_enrollment, na.rm = TRUE),
      share_students_with_absence = safe_divide(
        sum(count_students_one_or_more_absences, na.rm = TRUE),
        eligible_cumulative_enrollment
      ),
      avg_days_absent_per_eligible = safe_divide(
        sum(total_days_absent, na.rm = TRUE),
        eligible_cumulative_enrollment
      ),
      .groups = "drop"
    ) |>
    mutate(definition = definition_label, .before = 1)
}

burden_group_sensitivity <- bind_rows(
  make_burden_sensitivity(
    affected_cutoff,
    "Affected: top 25%",
    "Top 25% of affected school-years"
  ),
  make_burden_sensitivity(

```

```

    top_10_cutoff,
    "Affected: top 10%",
    "Top 10% of affected school-years"
  )
)

burden_group_sensitivity |>
  arrange(definition, sensitivity_group)

```

```

# A tibble: 6 x 8
  definition      sensitivity_group school_year_rows schools wildfire_student_days
  <chr>          <chr>                <int>    <int>                <dbl>
1 Top 10% of a~ Affected: below ~         2987     2501                2886488
2 Top 10% of a~ Affected: top 10%           332       322                2395724
3 Top 10% of a~ No wildfire        16548     9457                   0
4 Top 25% of a~ Affected: below ~         2489     2133                1631238
5 Top 25% of a~ Affected: top 25%          830       780                3650974
6 Top 25% of a~ No wildfire        16548     9457                   0
# i 3 more variables: eligible_cumulative_enrollment <dbl>,
#   share_students_with_absence <dbl>, avg_days_absent_per_eligible <dbl>

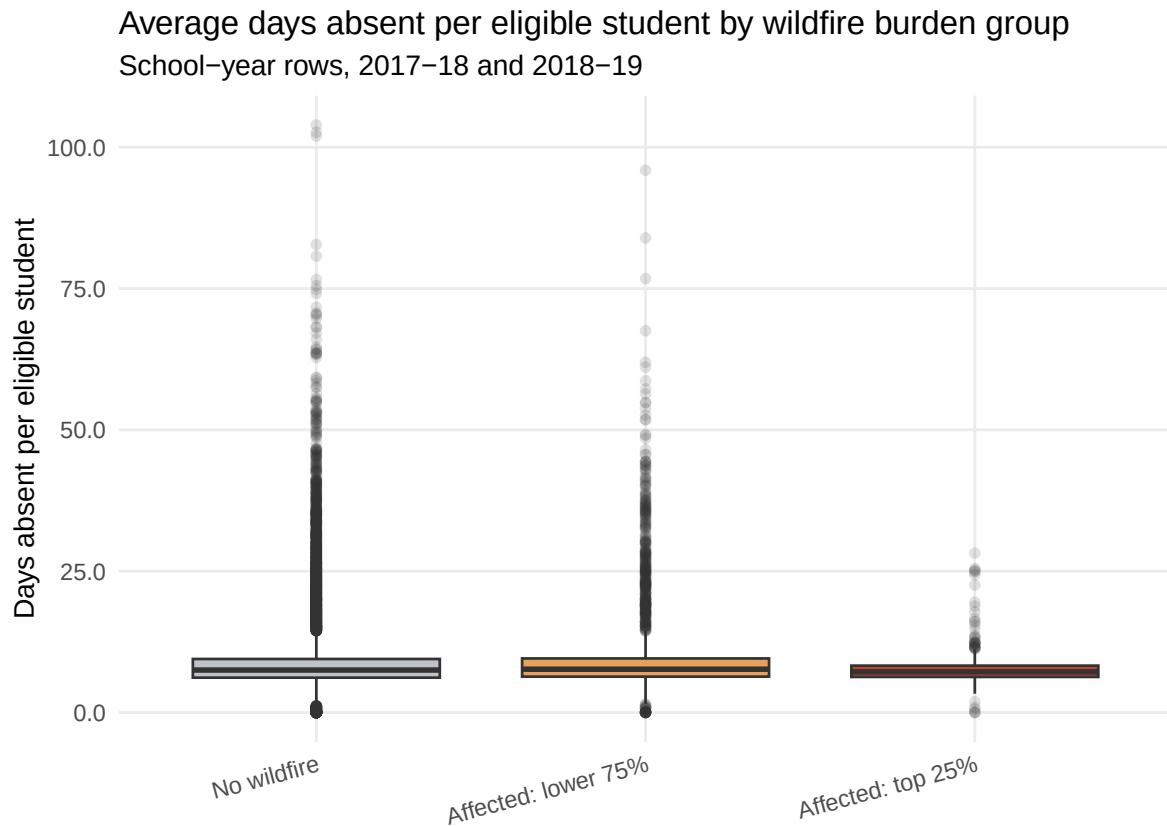
```

The sensitivity table checks whether the descriptive comparison depends entirely on the top-25% cutoff. It should be read cautiously: the highest wildfire-burden group is not automatically expected to have the highest absenteeism value because wildfire burden and absenteeism both vary with school, district, and community context.

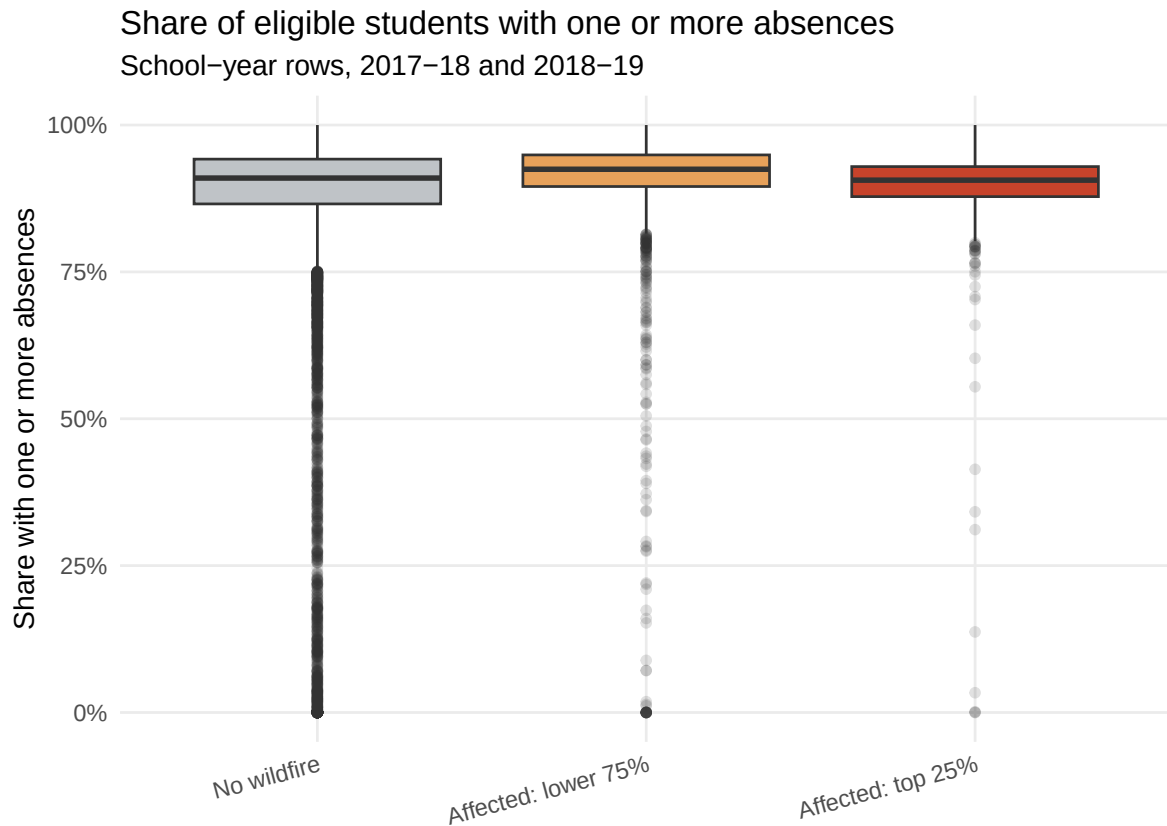
```

closure_absenteeism_joined |>
  filter(!is.na(avg_days_absent_per_eligible)) |>
  ggplot(aes(x = burden_group, y = avg_days_absent_per_eligible, fill = burden_group)) +
  geom_boxplot(outlier.alpha = 0.15) +
  scale_fill_manual(values = burden_palette, guide = "none") +
  scale_y_continuous(labels = label_number(accuracy = 0.1)) +
  labs(
    title = "Average days absent per eligible student by wildfire burden group",
    subtitle = "School-year rows, 2017-18 and 2018-19",
    x = NULL,
    y = "Days absent per eligible student"
  ) +
  theme(
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 15, hjust = 1)
  )

```



```
closure_absenteeism_joined |>
  filter(!is.na(share_students_with_absence)) |>
  ggplot(aes(x = burden_group, y = share_students_with_absence, fill = burden_group)) +
  geom_boxplot(outlier.alpha = 0.15) +
  scale_fill_manual(values = burden_palette, guide = "none") +
  scale_y_continuous(labels = label_percent()) +
  labs(
    title = "Share of eligible students with one or more absences",
    subtitle = "School-year rows, 2017-18 and 2018-19",
    x = NULL,
    y = "Share with one or more absences"
  ) +
  theme(
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 15, hjust = 1)
  )
```



```
reason_labels <- c(
  "excused_absences_count" = "Excused",
  "unexcused_absences_count" = "Unexcused",
  "out_of_school_suspension_absences_count" = "Suspension",
  "incomplete_independent_study_absences_count" = "Incomplete independent study"
)

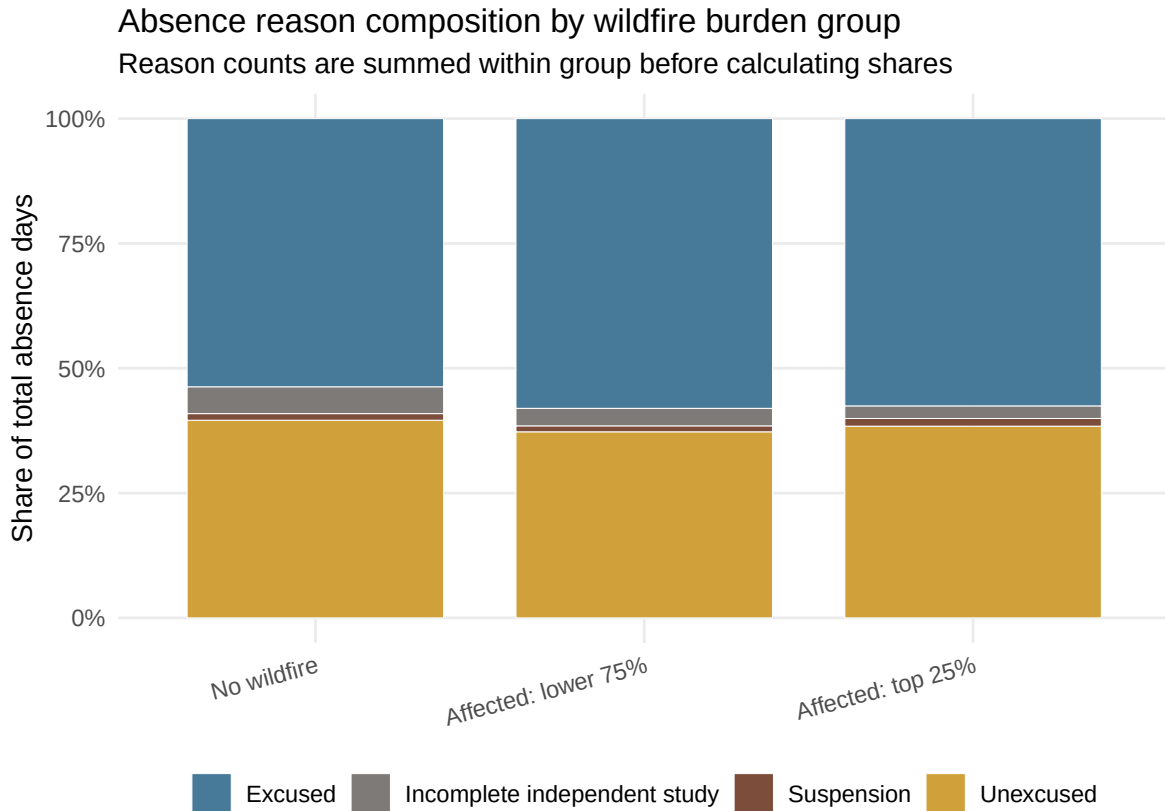
reason_composition_by_group <- closure_absenteeism_joined |>
  filter(!is.na(total_days_absent), total_days_absent > 0) |>
  group_by(burden_group) |>
  summarise(
    total_days_absent = sum(total_days_absent, na.rm = TRUE),
    across(all_of(absence_reason_count_vars), ~ sum(.x, na.rm = TRUE)),
    .groups = "drop"
  ) |>
  pivot_longer(
    cols = all_of(absence_reason_count_vars),
    names_to = "reason",
```

```

    values_to = "absence_days"
  ) |>
  mutate(
    reason = recode(reason, !!!reason_labels),
    reason_share = safe_divide(absence_days, total_days_absent)
  )

ggplot(
  reason_composition_by_group,
  aes(x = burden_group, y = reason_share, fill = reason)
) +
  geom_col(width = 0.78, color = "white", linewidth = 0.2) +
  scale_y_continuous(labels = label_percent()) +
  scale_fill_manual(
    values = c(
      "Excused" = "#477998",
      "Unexcused" = "#D0A03A",
      "Suspension" = "#7C4D3A",
      "Incomplete independent study" = "#7D7A78"
    ),
    name = NULL
  ) +
  labs(
    title = "Absence reason composition by wildfire burden group",
    subtitle = "Reason counts are summed within group before calculating shares",
    x = NULL,
    y = "Share of total absence days"
  ) +
  theme(
    legend.position = "bottom",
    panel.grid.minor = element_blank(),
    axis.text.x = element_text(angle = 15, hjust = 1)
  )

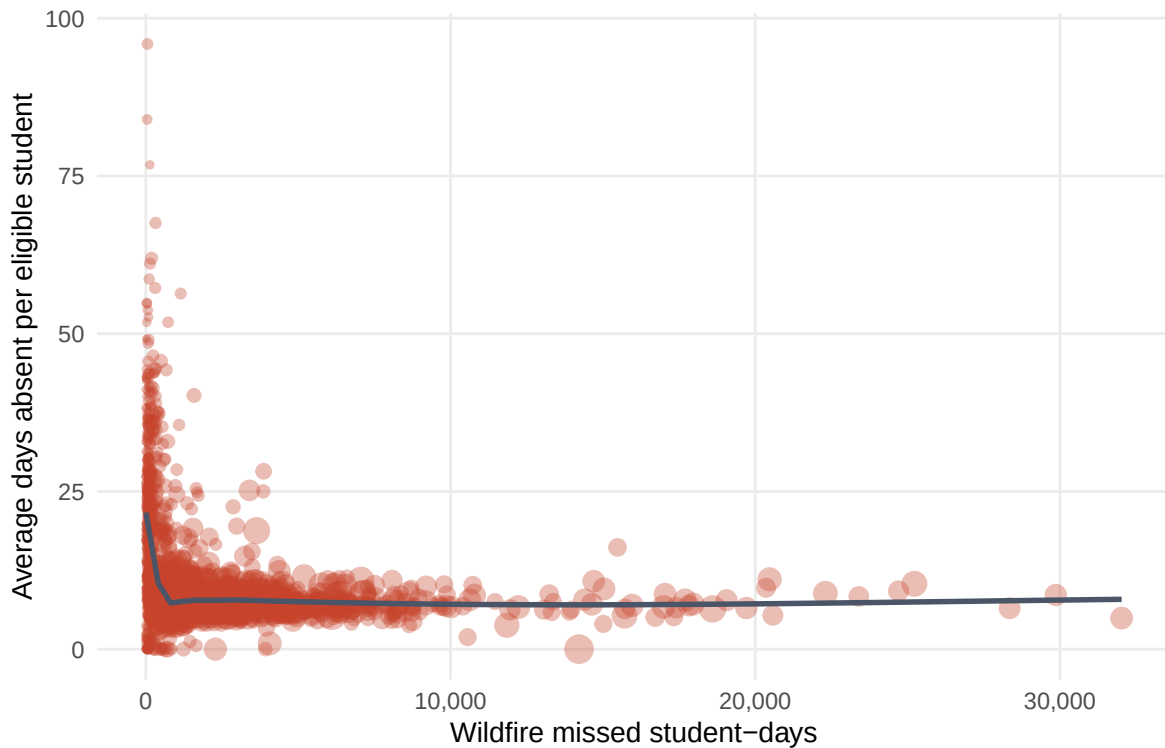
```



```
closure_absenteeism_joined |>
  filter(wildfire_student_days > 0, !is.na(avg_days_absent_per_eligible)) |>
  ggplot(aes(x = wildfire_student_days, y = avg_days_absent_per_eligible)) +
  geom_point(aes(size = eligible_cumulative_enrollment), alpha = 0.35, color = "#C7432B") +
  geom_smooth(method = "loess", se = FALSE, color = "#4A5568") +
  scale_x_continuous(labels = label_comma()) +
  scale_size_continuous(labels = label_comma(), guide = "none") +
  labs(
    title = "Wildfire burden and absenteeism among affected school-years",
    subtitle = "Descriptive only; this is not a causal estimate",
    x = "Wildfire missed student-days",
    y = "Average days absent per eligible student"
  ) +
  theme(panel.grid.minor = element_blank())
```

Wildfire burden and absenteeism among affected school-years

Descriptive only; this is not a causal estimate



County-level absence-day bridge (Absenteeism by Reason)

```
county_absenteeism_bridge <- closure_absenteeism_joined |>
  filter(!is.na(avg_days_absent_per_eligible), !is.na(county), county != "") |>
  group_by(county) |>
  summarise(
    school_year_rows = n(),
    schools = n_distinct(cds_code),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    eligible_cumulative_enrollment = sum(eligible_cumulative_enrollment, na.rm = TRUE),
    count_students_one_or_more_absences = sum(count_students_one_or_more_absences, na.rm = TRUE),
    total_days_absent = sum(total_days_absent, na.rm = TRUE),
    share_students_with_absence = safe_divide(
      count_students_one_or_more_absences,
      eligible_cumulative_enrollment
    )
  )
```

```

    ),
    avg_days_absent_per_eligible = safe_divide(
      total_days_absent,
      eligible_cumulative_enrollment
    ),
    .groups = "drop"
  ) |>
  mutate(any_wildfire_burden = wildfire_student_days > 0)

county_absenteeism_bridge |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    school_year_rows,
    schools,
    wildfire_days,
    wildfire_student_days,
    eligible_cumulative_enrollment,
    avg_days_absent_per_eligible,
    share_students_with_absence
  ) |>
  slice_head(n = 15)

```

A tibble: 15 x 8

	county	school_year_rows	schools	wildfire_days	wildfire_student_days
	<chr>	<int>	<int>	<dbl>	<dbl>
1	Ventura	414	209	1540	1052437
2	Sonoma	352	180	1768	755293
3	Los Angeles	4445	2252	876	635624
4	Riverside	978	497	366	362906
5	Contra Costa	535	276	584	355246
6	Butte	176	91	907	324435
7	Santa Barbara	239	125	563	271888
8	Napa	84	44	447	219721
9	Sacramento	761	383	304	209112
10	Solano	205	104	251	163045
11	Alameda	763	388	178	129436
12	San Diego	1512	775	166	111195
13	Lake	75	38	339	88738
14	Marin	146	73	214	86485
15	Orange	1219	613	107	78093

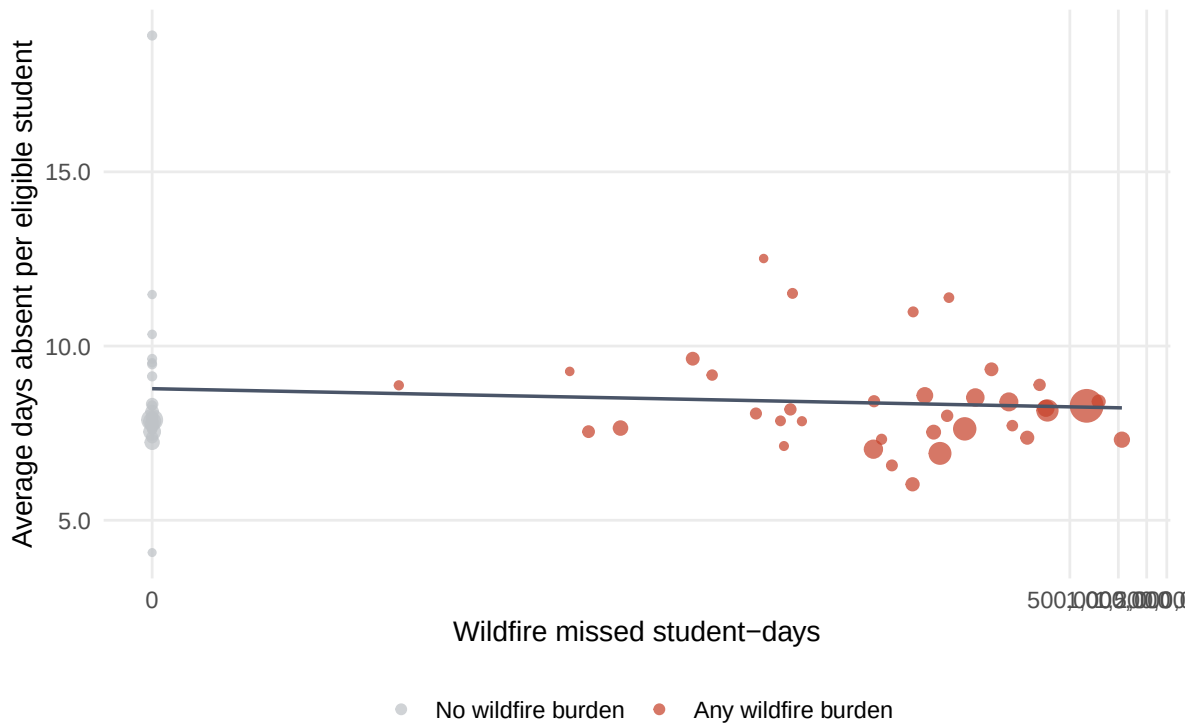
i 3 more variables: eligible_cumulative_enrollment <dbl>,


```
# avg_days_absent_per_eligible <dbl>, share_students_with_absence <dbl>
```

```
ggplot(  
  county_absenteeism_bridge,  
  aes(x = wildfire_student_days, y = avg_days_absent_per_eligible)  
) +  
  geom_point(  
    aes(size = eligible_cumulative_enrollment, color = any_wildfire_burden),  
    alpha = 0.72  
  ) +  
  geom_smooth(method = "lm", se = FALSE, color = "#4A5568", linewidth = 0.7) +  
  scale_x_continuous(  
    trans = pseudo_log_trans(base = 10),  
    labels = label_comma()  
  ) +  
  scale_y_continuous(labels = label_number(accuracy = 0.1)) +  
  scale_size_continuous(labels = label_comma(), guide = "none") +  
  scale_color_manual(  
    values = c("FALSE" = "#BFC3C7", "TRUE" = "#C7432B"),  
    labels = c("No wildfire burden", "Any wildfire burden"),  
    name = NULL  
  ) +  
  labs(  
    title = "County wildfire burden and absenteeism patterns",  
    subtitle = "County summaries aggregate matched school-year rows; descriptive only",  
    x = "Wildfire missed student-days",  
    y = "Average days absent per eligible student"  
  ) +  
  theme(  
    legend.position = "bottom",  
    panel.grid.minor = element_blank()  
  )  
)
```

County wildfire burden and absenteeism patterns

County summaries aggregate matched school-year rows; descriptive only



This county view supplements the chronic absenteeism county analysis with absence-day measures from the Absenteeism by Reason files. It does not estimate the effect of wildfire closures on absence days; counties differ in many ways that are not controlled here.

High-burden school profiles

```
school_absenteeism_profile <- closure_absenteeism_joined |>
  group_by(cds_code, county, district_name, school_site) |>
  summarise(
    years_observed = n_distinct(closure_year),
    wildfire_days = sum(days_wildfire, na.rm = TRUE),
    days_total = sum(days_total, na.rm = TRUE),
    wildfire_student_days = sum(wildfire_student_days, na.rm = TRUE),
    total_enrollment_school_years = sum(total_enrollment, na.rm = TRUE),
    eligible_cumulative_enrollment = sum(eligible_cumulative_enrollment, na.rm = TRUE),
    count_students_one_or_more_absences = sum(count_students_one_or_more_absences, na.rm = TRUE),
    total_days_absent = sum(total_days_absent, na.rm = TRUE),
```

```

    across(all_of(absence_reason_count_vars), ~ sum(.x, na.rm = TRUE)),
    .groups = "drop"
  ) |>
  mutate(
    wildfire_share_of_total = safe_divide(wildfire_days, days_total),
    avg_days_absent_per_eligible = safe_divide(total_days_absent, eligible_cumulative_enrollm
    share_students_with_absence = safe_divide(
      count_students_one_or_more_absences,
      eligible_cumulative_enrollment
    ),
    avg_days_absent_among_absent = safe_divide(
      total_days_absent,
      count_students_one_or_more_absences
    ),
    excused_absence_share = safe_divide(excused_absences_count, total_days_absent),
    unexcused_absence_share = safe_divide(unexcused_absences_count, total_days_absent),
    suspension_absence_share = safe_divide(out_of_school_suspension_absences_count, total_da
    incomplete_independent_study_absence_share = safe_divide(
      incomplete_independent_study_absences_count,
      total_days_absent
    )
  )
)

top_school_absenteeism_profiles <- school_absenteeism_profile |>
  filter(wildfire_student_days > 0) |>
  arrange(desc(wildfire_student_days)) |>
  select(
    county,
    district_name,
    school_site,
    years_observed,
    wildfire_days,
    wildfire_student_days,
    eligible_cumulative_enrollment,
    avg_days_absent_per_eligible,
    share_students_with_absence,
    excused_absence_share,
    unexcused_absence_share,
    suspension_absence_share,
    incomplete_independent_study_absence_share
  ) |>
  slice_head(n = 25)

```

top_school_absenteeism_profiles

A tibble: 25 x 13

	county	district_name	school_site	years_observed	wildfire_days
	<chr>	<chr>	<chr>	<int>	<dbl>
1	Sonoma	Santa Rosa High	Santa Rosa High	2	17
2	Ventura	Ventura Unified	Ventura High	2	15
3	Ventura	Oxnard Union High	Pacifica High	2	10
4	Ventura	Ventura Unified	Buena High	2	15
5	Sonoma	Santa Rosa High	Montgomery High	2	17
6	Ventura	Oxnard Union High	Oxnard High	2	10
7	Sonoma	Santa Rosa High	Maria Carrillo High	2	17
8	Ventura	Oxnard Union High	Channel Islands High	2	10
9	Sonoma	Santa Rosa High	Piner High	2	17
10	Ventura	Oxnard Union High	Adolfo Camarillo High	2	10

i 15 more rows

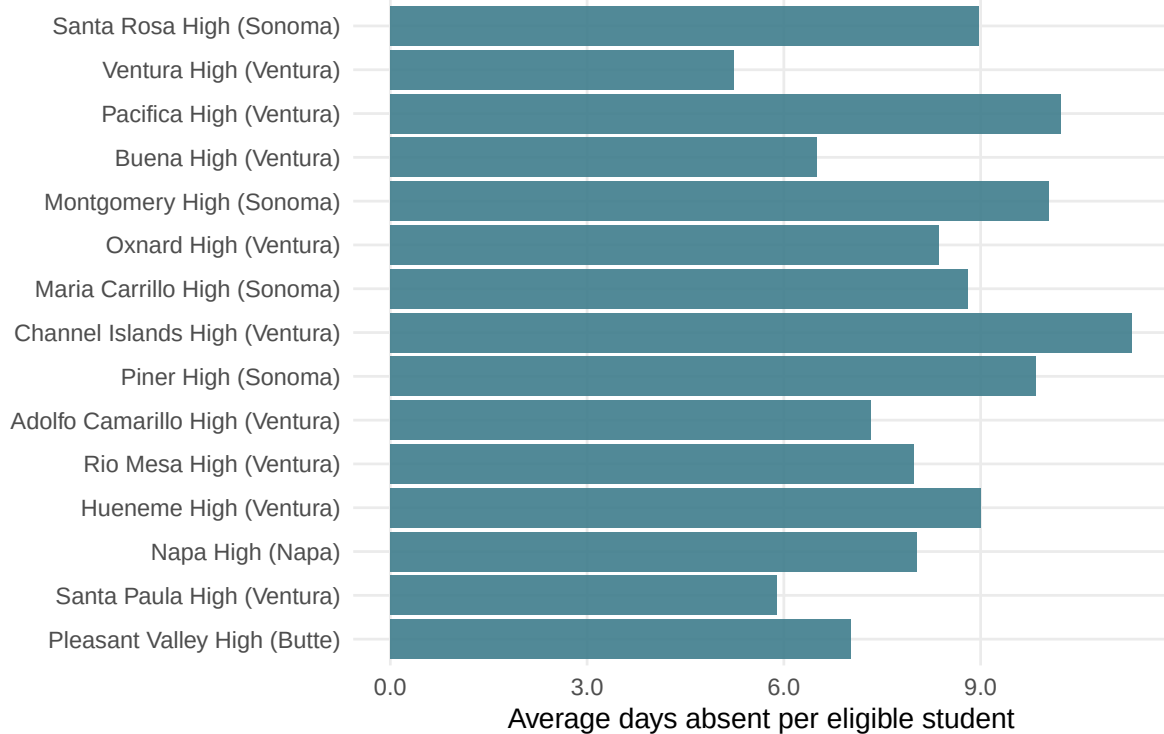
i 8 more variables: wildfire_student_days <dbl>,
eligible_cumulative_enrollment <dbl>, avg_days_absent_per_eligible <dbl>,
share_students_with_absence <dbl>, excused_absence_share <dbl>,
unexcused_absence_share <dbl>, suspension_absence_share <dbl>,
incomplete_independent_study_absence_share <dbl>

```
top_school_plot_data <- school_absenteeism_profile |>
  filter(wildfire_student_days > 0) |>
  slice_max(wildfire_student_days, n = 15, with_ties = FALSE) |>
  mutate(school_label = str_c(str_trunc(school_site, 38), " (", county, ")"))

ggplot(
  top_school_plot_data,
  aes(x = avg_days_absent_per_eligible, y = fct_reorder(school_label, wildfire_student_days))
) +
  geom_col(fill = "#3F7D8C", alpha = 0.9) +
  scale_x_continuous(labels = label_number(accuracy = 0.1)) +
  labs(
    title = "Absenteeism in the highest wildfire-burden schools",
    subtitle = "Top schools ranked by wildfire missed student-days, 2017-18 and 2018-19",
    x = "Average days absent per eligible student",
    y = NULL
  ) +
  theme(panel.grid.minor = element_blank())
```

Absenteeism in the highest wildfire-burden schools

Top schools ranked by wildfire missed student-days, 2017-18 and



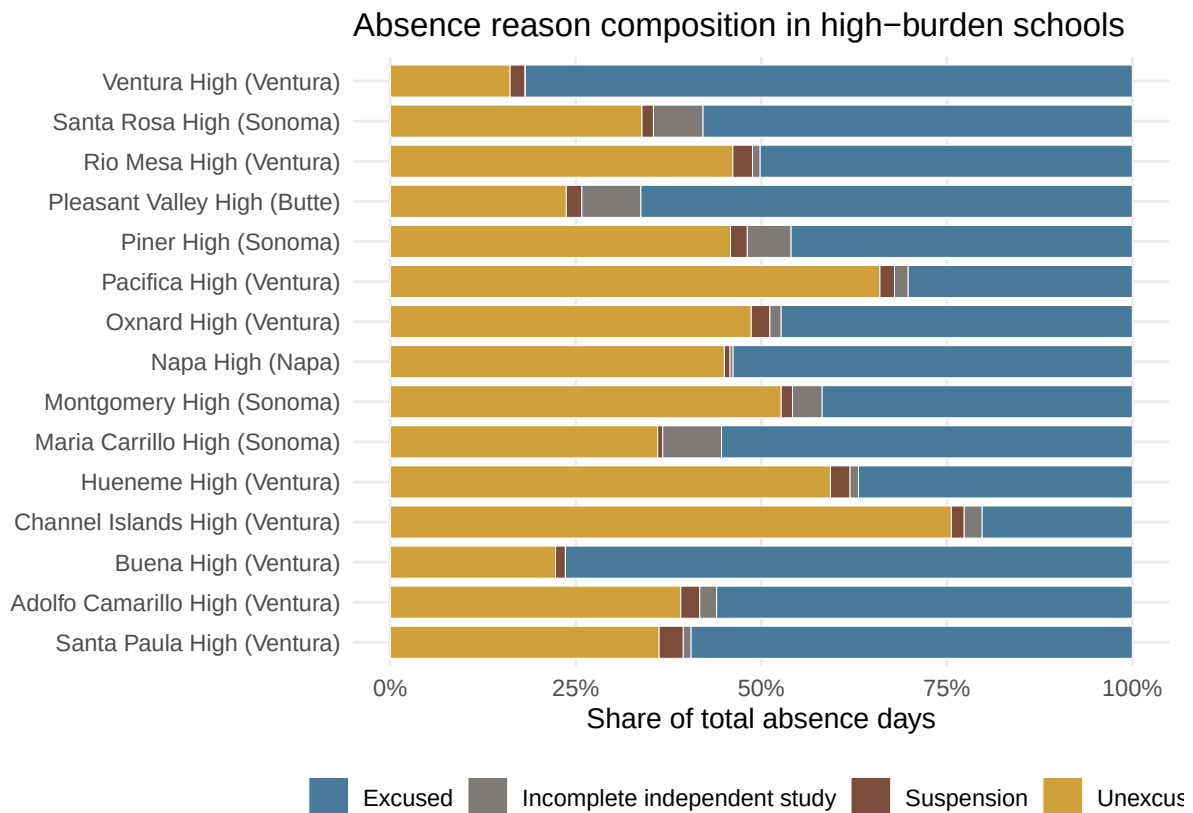
```
top_school_reason_long <- top_school_plot_data |>
  select(school_label, total_days_absent, all_of(absence_reason_count_vars)) |>
  pivot_longer(
    cols = all_of(absence_reason_count_vars),
    names_to = "reason",
    values_to = "absence_days"
  ) |>
  mutate(
    reason = recode(reason, !!!reason_labels),
    reason_share = safe_divide(absence_days, total_days_absent)
  )

ggplot(
  top_school_reason_long,
  aes(x = reason_share, y = fct_reorder(school_label, reason_share, .fun = sum), fill = reason)
) +
  geom_col(width = 0.8, color = "white", linewidth = 0.15) +
  scale_x_continuous(labels = label_percent()) +
```

```

scale_fill_manual(
  values = c(
    "Excused" = "#477998",
    "Unexcused" = "#D0A03A",
    "Suspension" = "#7C4D3A",
    "Incomplete independent study" = "#7D7A78"
  ),
  name = NULL
) +
labs(
  title = "Absence reason composition in high-burden schools",
  x = "Share of total absence days",
  y = NULL
) +
theme(
  legend.position = "bottom",
  panel.grid.minor = element_blank()
)

```



Step 5 takeaway

Matched absenteeism records show descriptive differences across wildfire burden groups and in high-burden school profiles. The pattern is not a causal estimate, and the most wildfire-burdened group is not necessarily the group with the highest average days absent. The safer interpretation is that wildfire burden, absence frequency, and absence reason composition vary together in ways that are worth describing, but the EDA does not isolate wildfire as the cause of absenteeism differences.

Appendix: closure burden and test outcomes

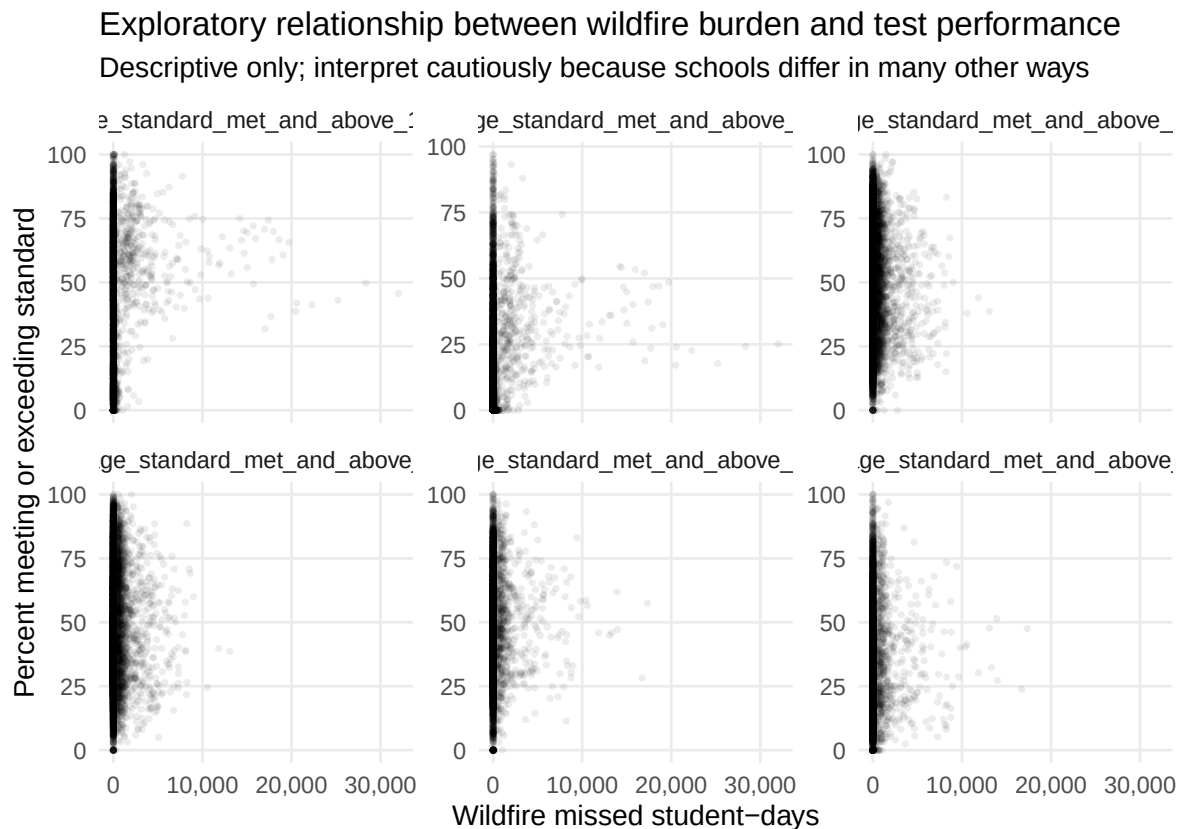
This appendix is retained as an exploratory check only. It is not part of the main wildfire-closure-to-absenteeism story and should not be treated as a causal model.

```
test_score_vars <- c(
  "percentage_standard_met_and_above_3_math",
  "percentage_standard_met_and_above_3_english",
  "percentage_standard_met_and_above_8_math",
  "percentage_standard_met_and_above_8_english",
  "percentage_standard_met_and_above_11_math",
  "percentage_standard_met_and_above_11_english"
)

test_score_long <- df |>
  filter(year %in% analysis_years) |>
  select(cds_code, year, county, days_wildfire, wildfire_student_days, any_of(test_score_vars)) |>
  mutate(across(any_of(test_score_vars), to_number)) |>
  pivot_longer(
    cols = any_of(test_score_vars),
    names_to = "test_metric",
    values_to = "percent_met_or_above"
  ) |>
  filter(!is.na(percent_met_or_above))

ggplot(
  test_score_long,
  aes(x = wildfire_student_days, y = percent_met_or_above)
) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.08, size = 0.7) +
  geom_smooth(method = "loess", se = FALSE, color = "#C7432B") +
  facet_wrap(vars(test_metric), scales = "free_y") +
  scale_x_continuous(labels = label_comma()) +
```

```
labs(
  title = "Exploratory relationship between wildfire burden and test performance",
  subtitle = "Descriptive only; interpret cautiously because schools differ in many other ways",
  x = "Wildfire missed student-days",
  y = "Percent meeting or exceeding standard"
) +
theme(panel.grid.minor = element_blank())
```



Limitations

This EDA is descriptive, not causal. It documents where wildfire closure burden appears in the available closure records, how county chronic absenteeism rates line up with that burden, and how matched absenteeism-by-reason measures look in affected school-years. It does not estimate what absenteeism would have been without wildfire closures.

The attendance analyses are limited to 2017-18 and 2018-19 because those CDE Chronic Absenteeism and Absenteeism by Reason years match closure year == 2018 and year == 2019.

The 2019-20 absenteeism files are excluded because CDE marks that year as unavailable/not valid due to COVID-era disruption.

CDE chronic absenteeism rates reflect CALPADS eligibility rules: students are excluded from the chronic absenteeism denominator when they are expected to attend fewer than 31 instructional days, enrolled but did not attend, or meet exempt-status conditions. Suppressed * values are treated as missing, not zero.

The closure-to-Absenteeism-by-Reason match rate is high but incomplete. Unmatched records mean the school-year absence-reason summaries cover most, but not all, closure school-year rows; the unmatched examples should be treated as diagnostics rather than a complete classification of missingness.

County aggregation can hide within-county school differences. A county can have high total wildfire missed student-days because it has large enrollment, while the most intense closure burden may be concentrated in a smaller set of schools or districts.

Finally, wildfire burden and absenteeism may both reflect other school and community differences, including baseline attendance patterns, grade configuration, district reporting context, socioeconomic conditions, displacement, smoke exposure, local health conditions, and other disruptions. These results are useful for describing patterns and motivating follow-up analysis, not for making causal claims.